

REPORTE DE EJECUCIÓN
ALGORITMO GENÉTICO EN ESPACIO 2D Y 3D

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MÉTODOS UTILIZADOS:

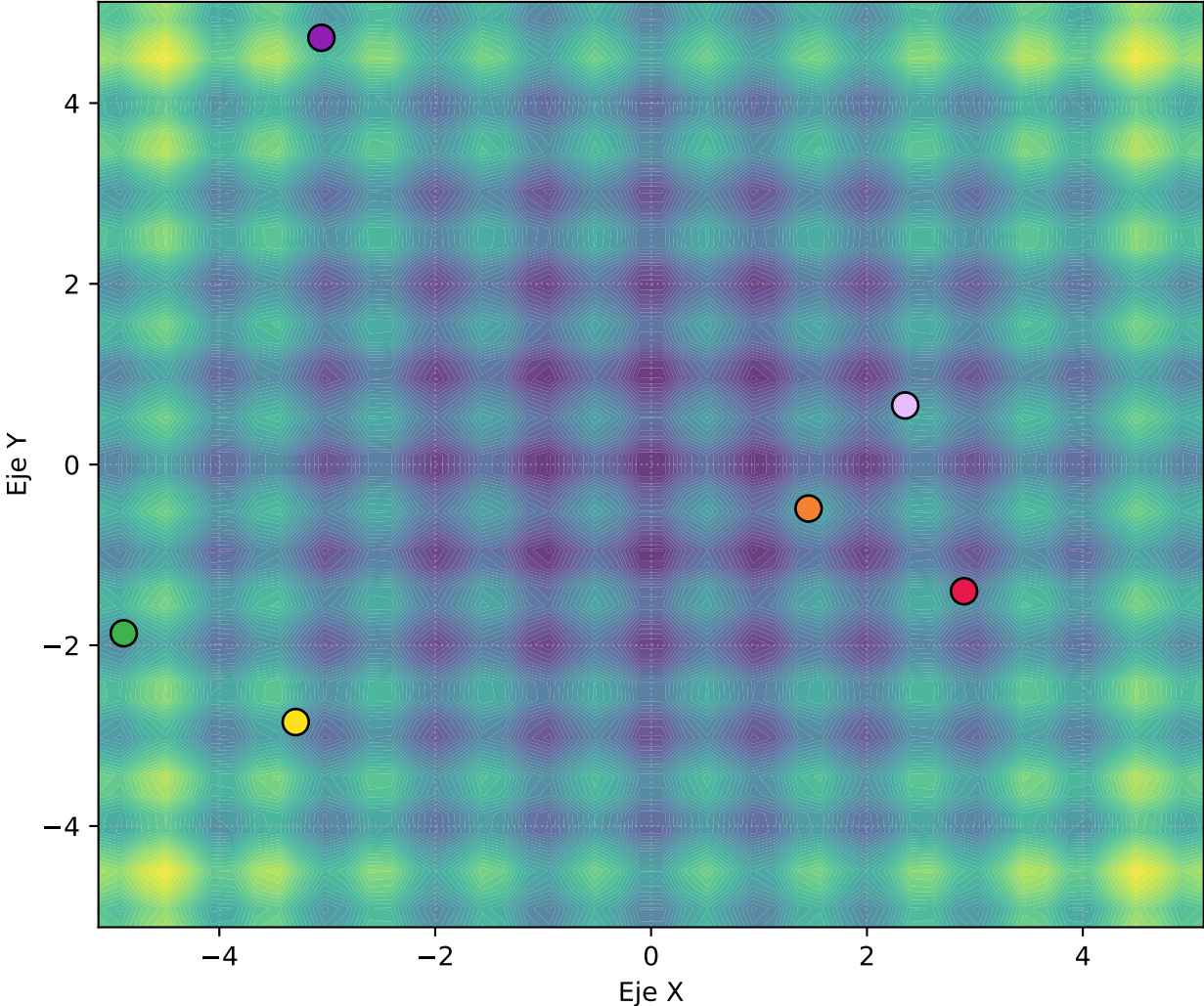
- > Aptitud 3D: □ RASTRIGIN
- > Codificación: □ BINARIO_2D
- > Selección: □ TORNEO
- > Cruza: □ UN_PUNTO
- > Mutación: □ UN_BIT
- > Generaciones: □ 100

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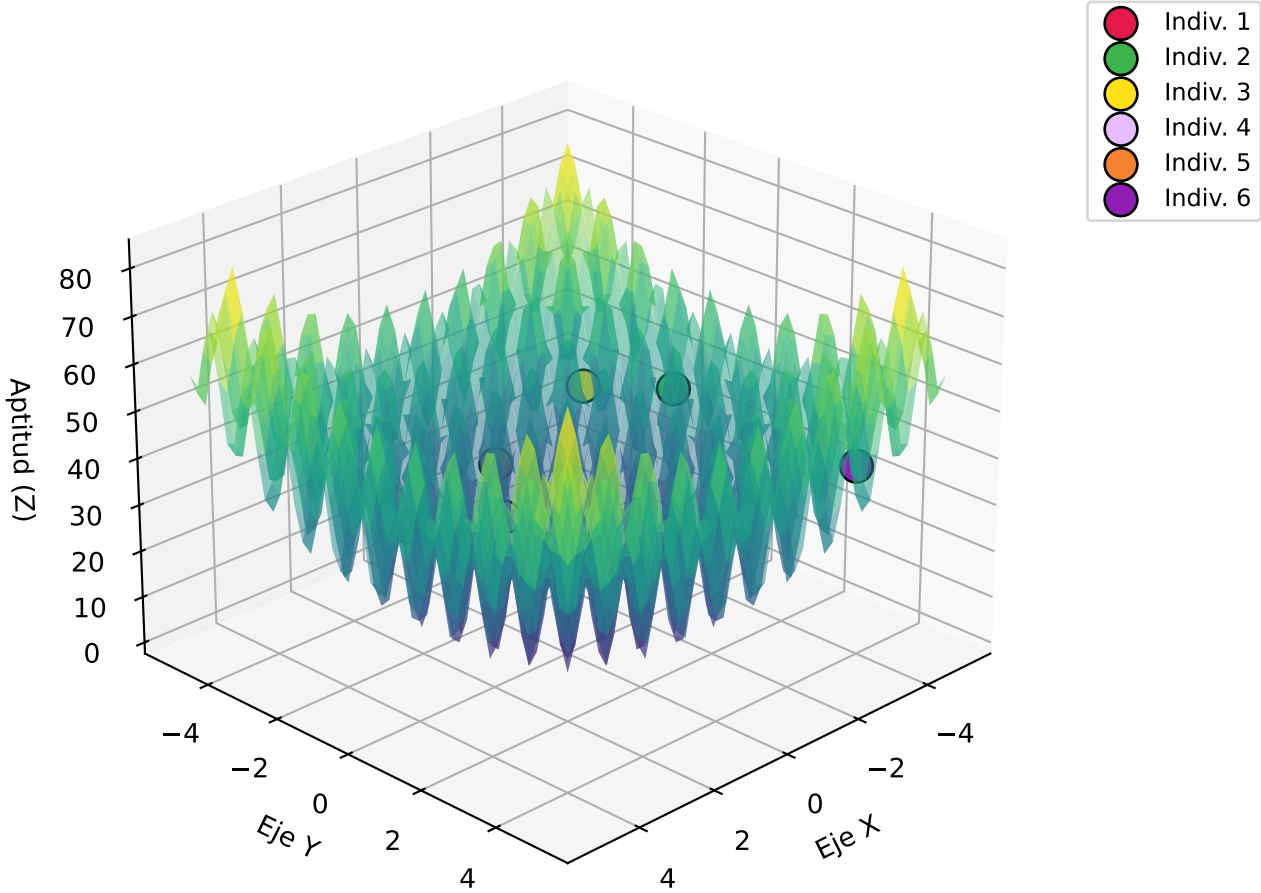
Centro de Investigaciones en Óptica

Generación 1: Posición Inicial de los 6 Mejores

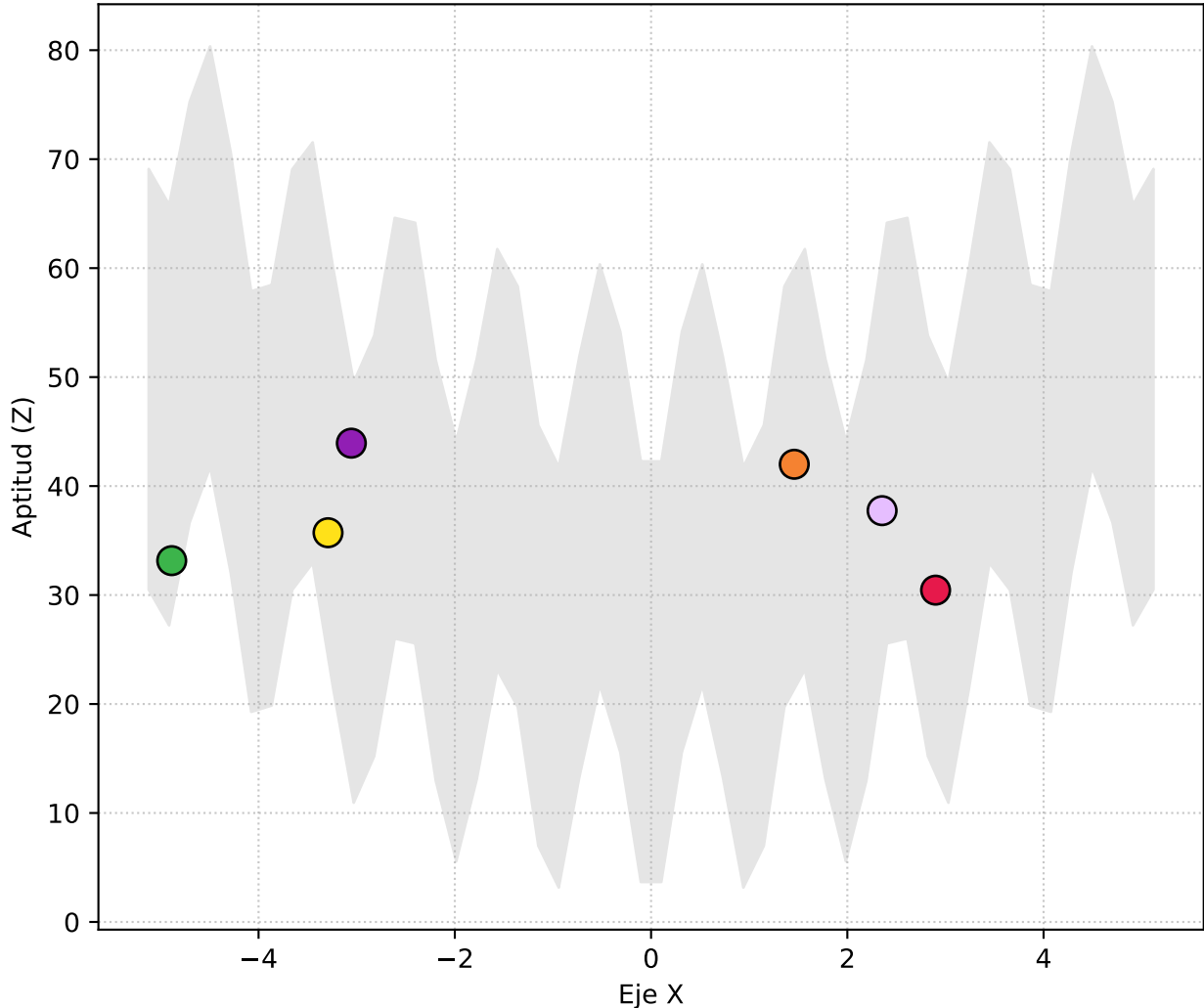
Vista Superior (X, Y)



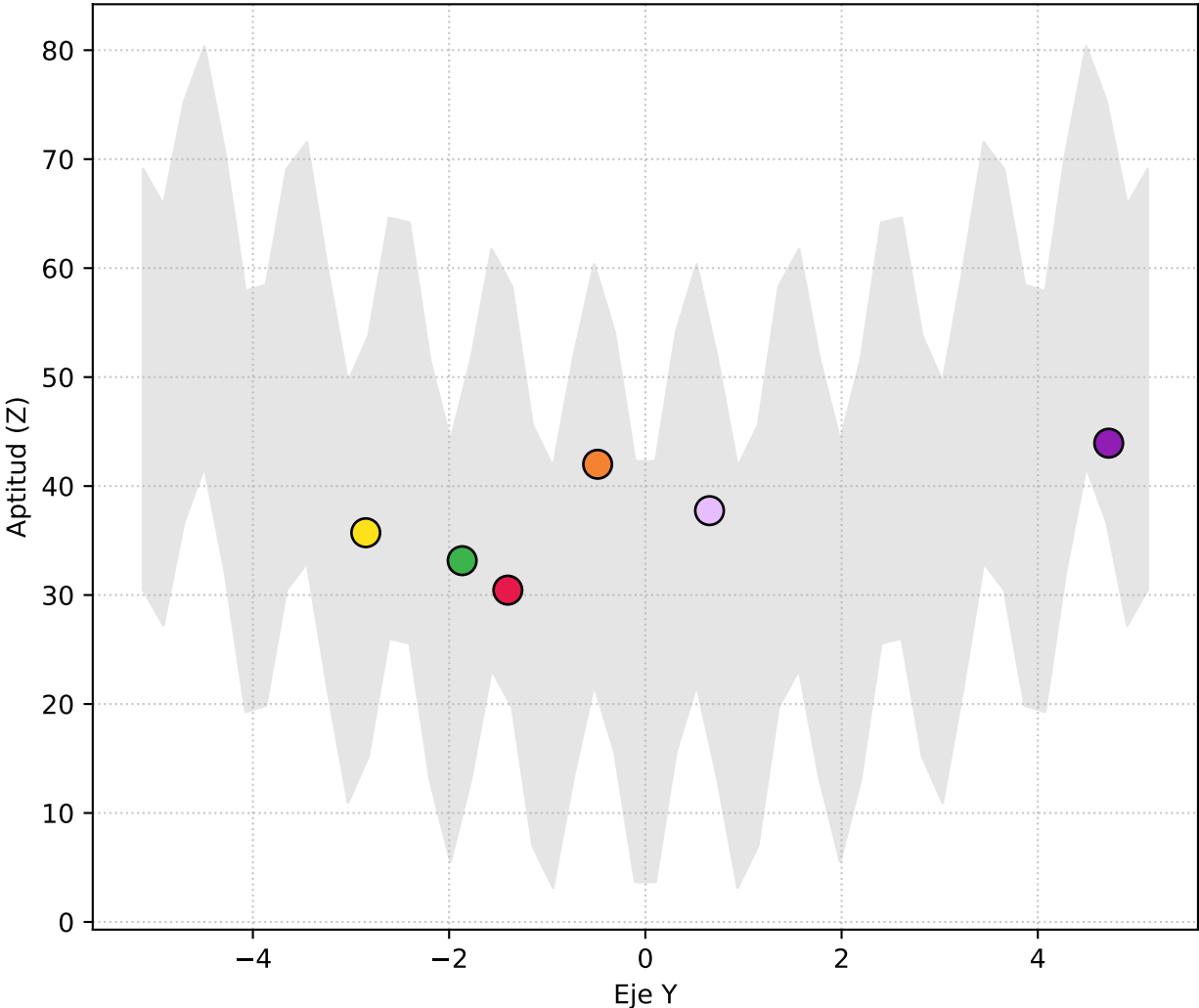
Perspectiva 3D



Vista Lateral (X vs Elevación Z)

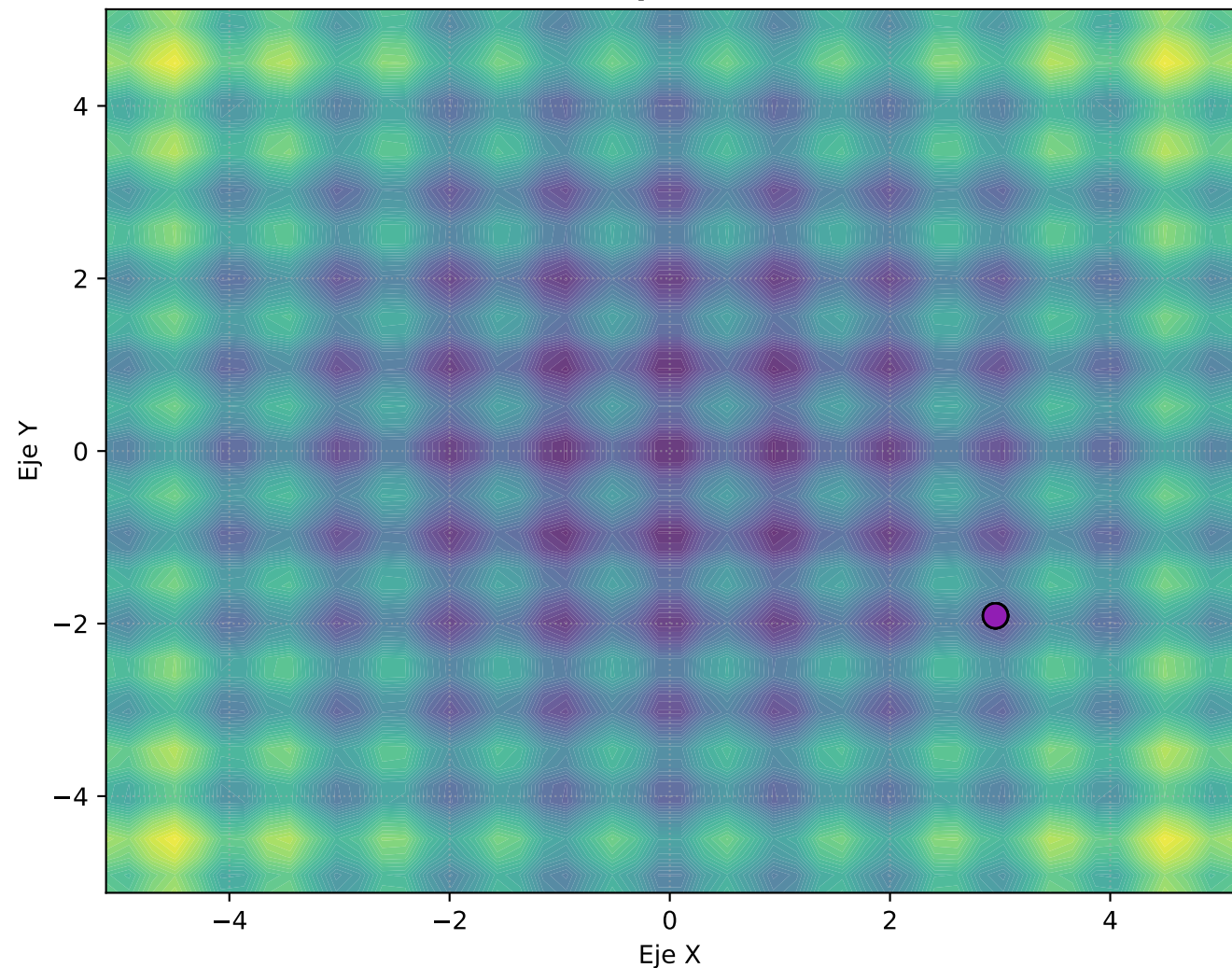


Vista Frontal (Y vs Elevación Z)

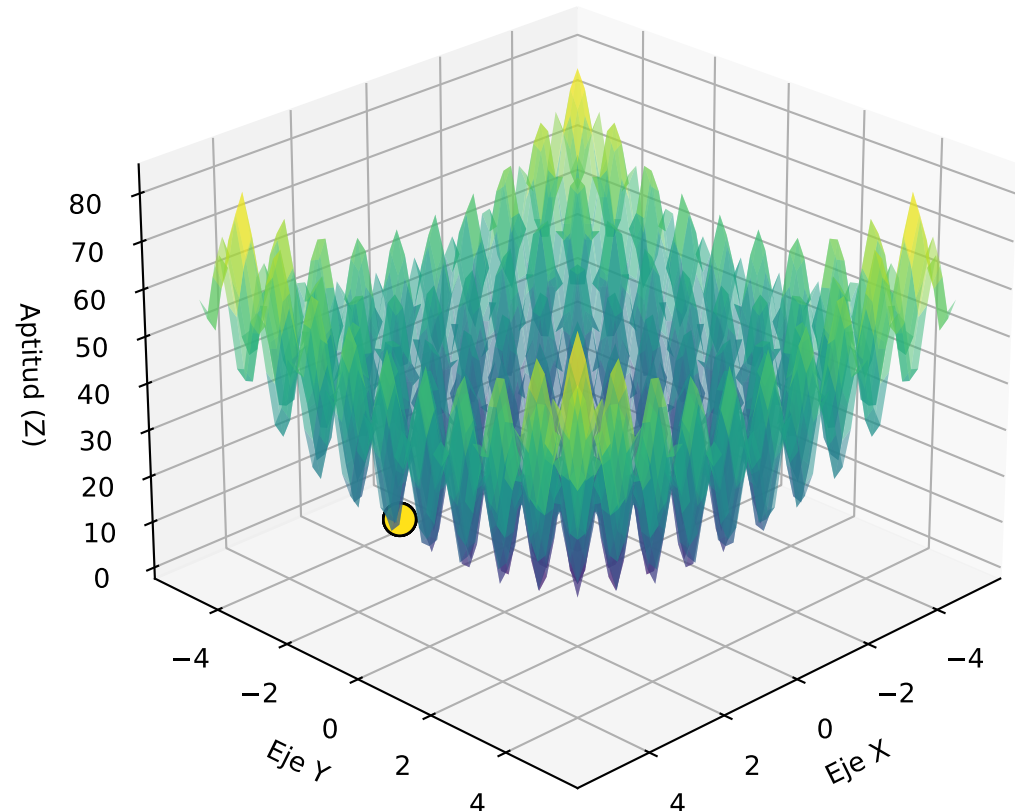


Progreso: Generación 10

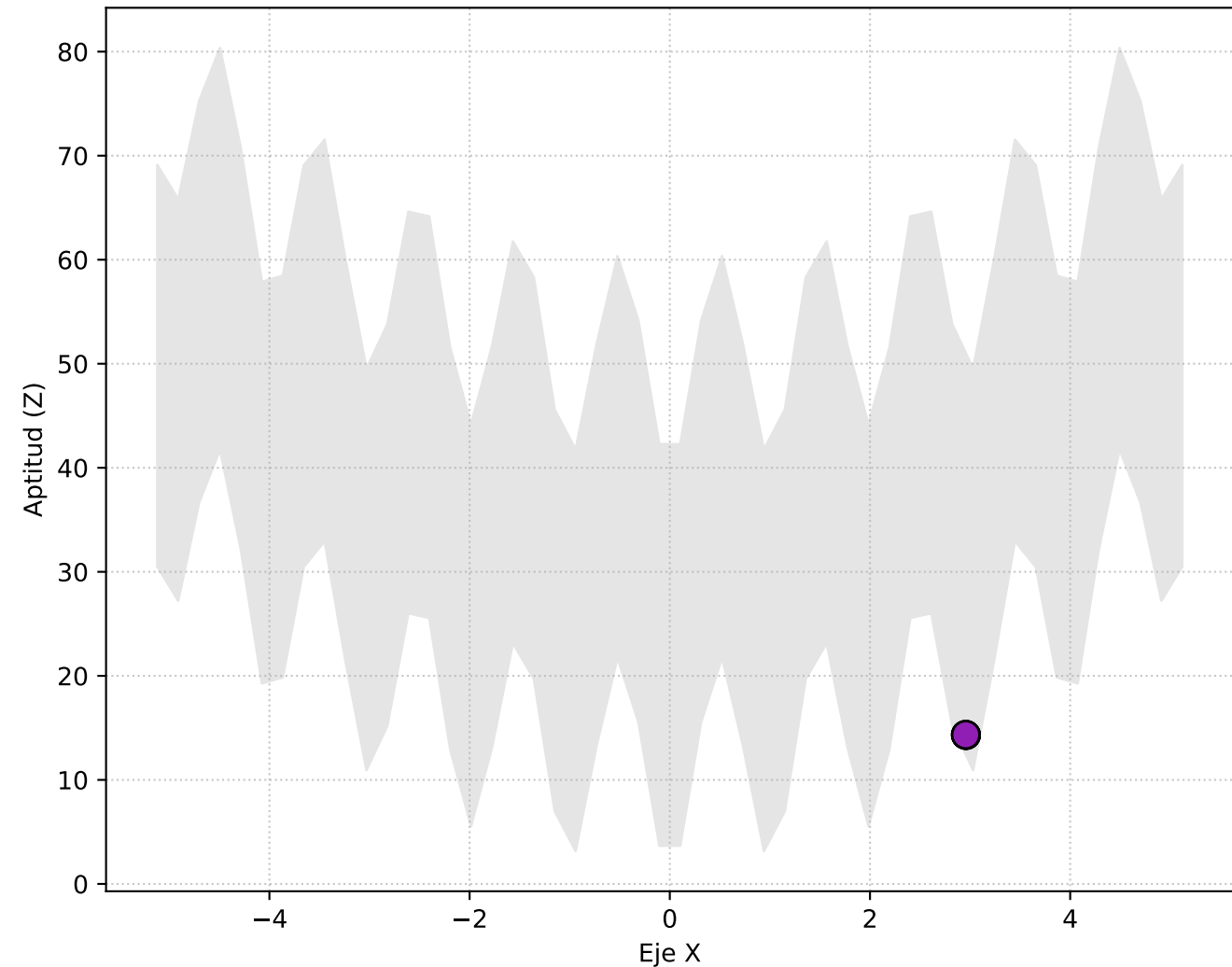
Vista Superior (X, Y)



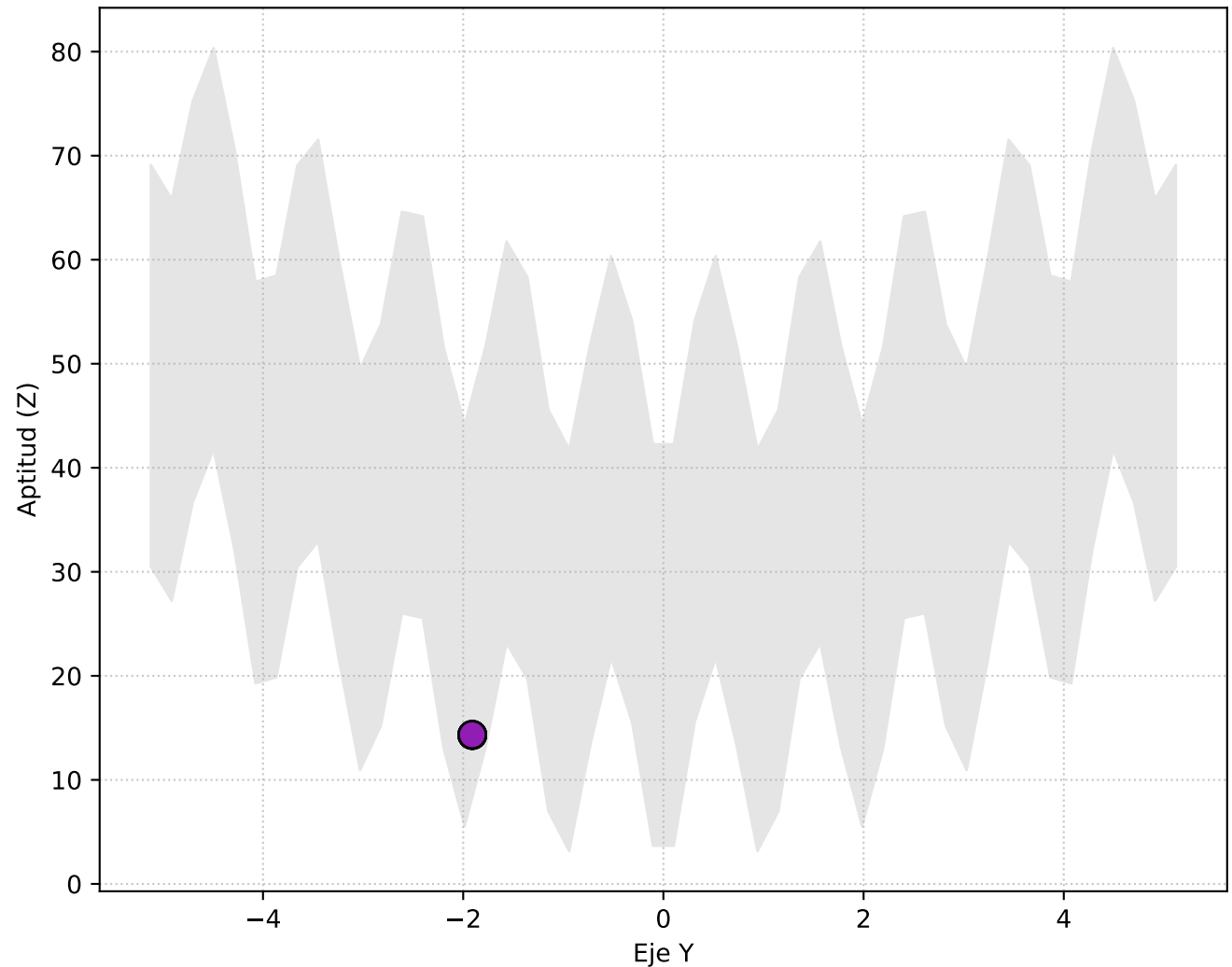
Perspectiva 3D



Vista Lateral (X vs Elevación Z)

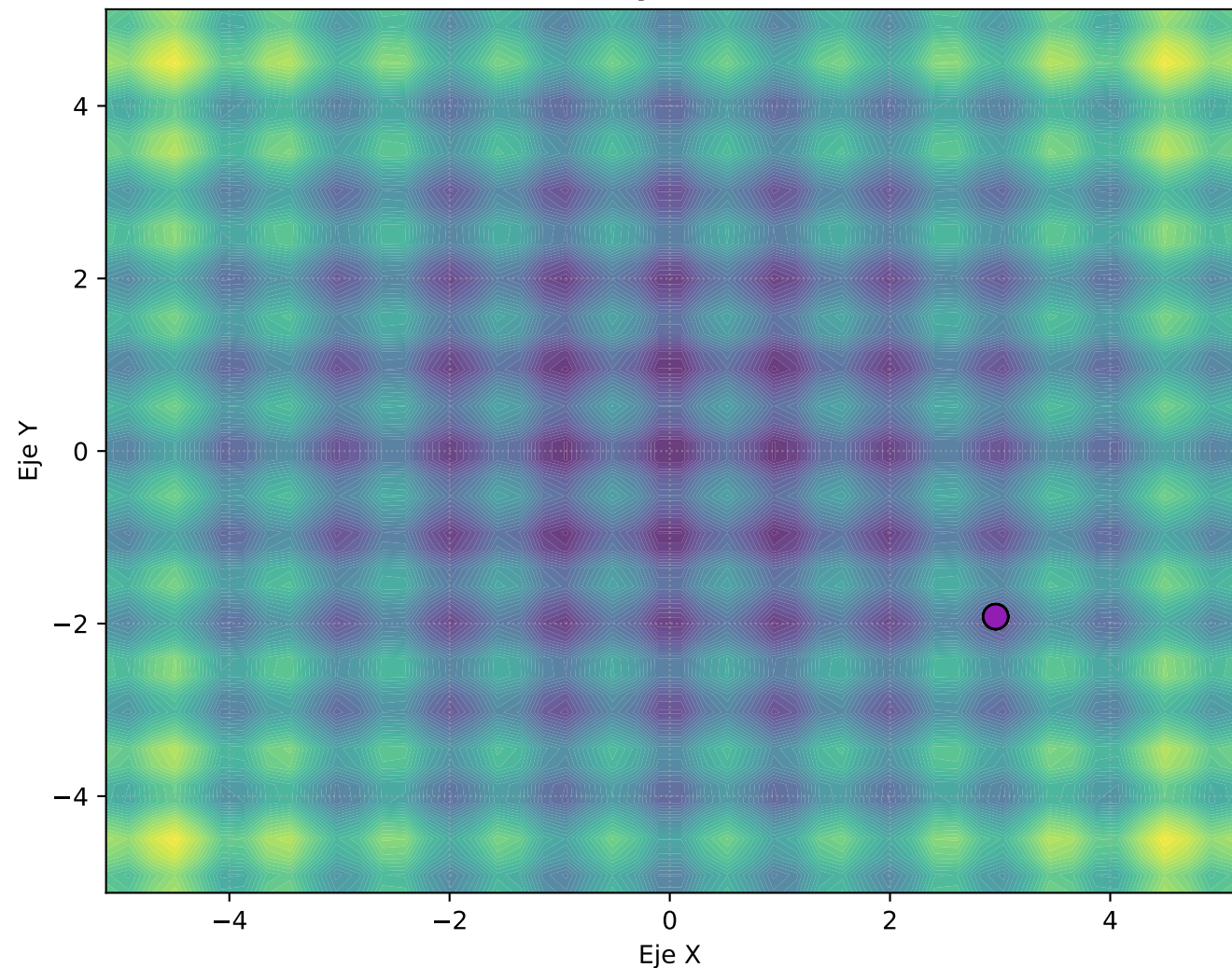


Vista Frontal (Y vs Elevación Z)

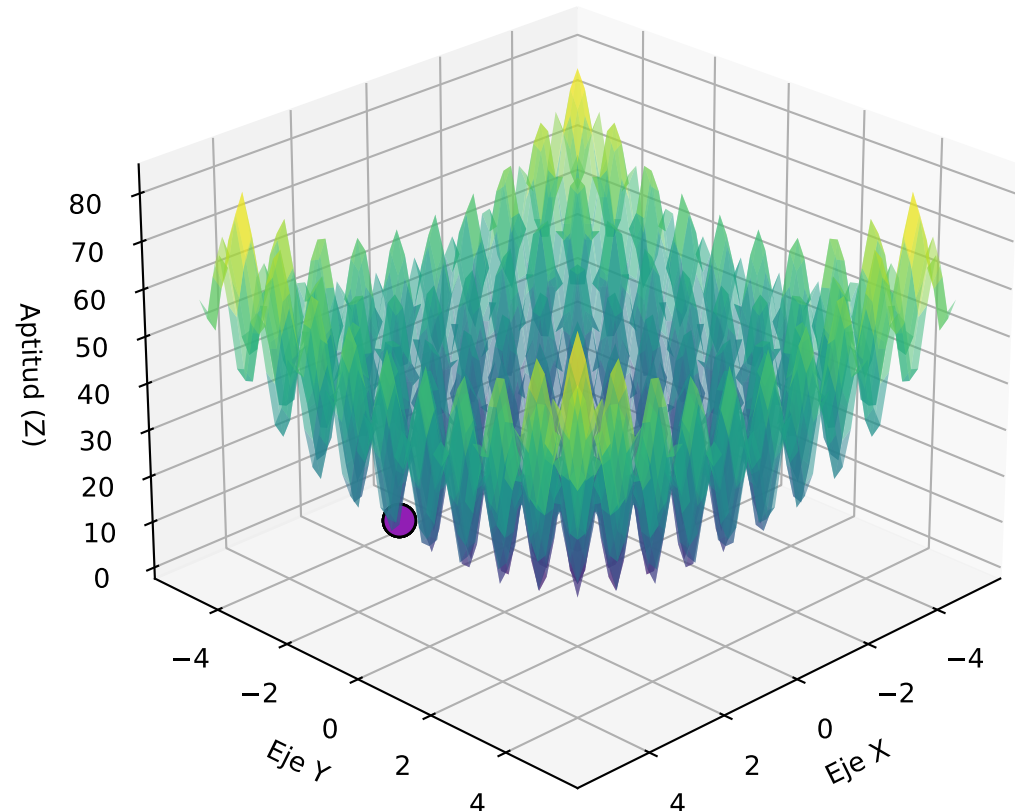


Progreso: Generación 20

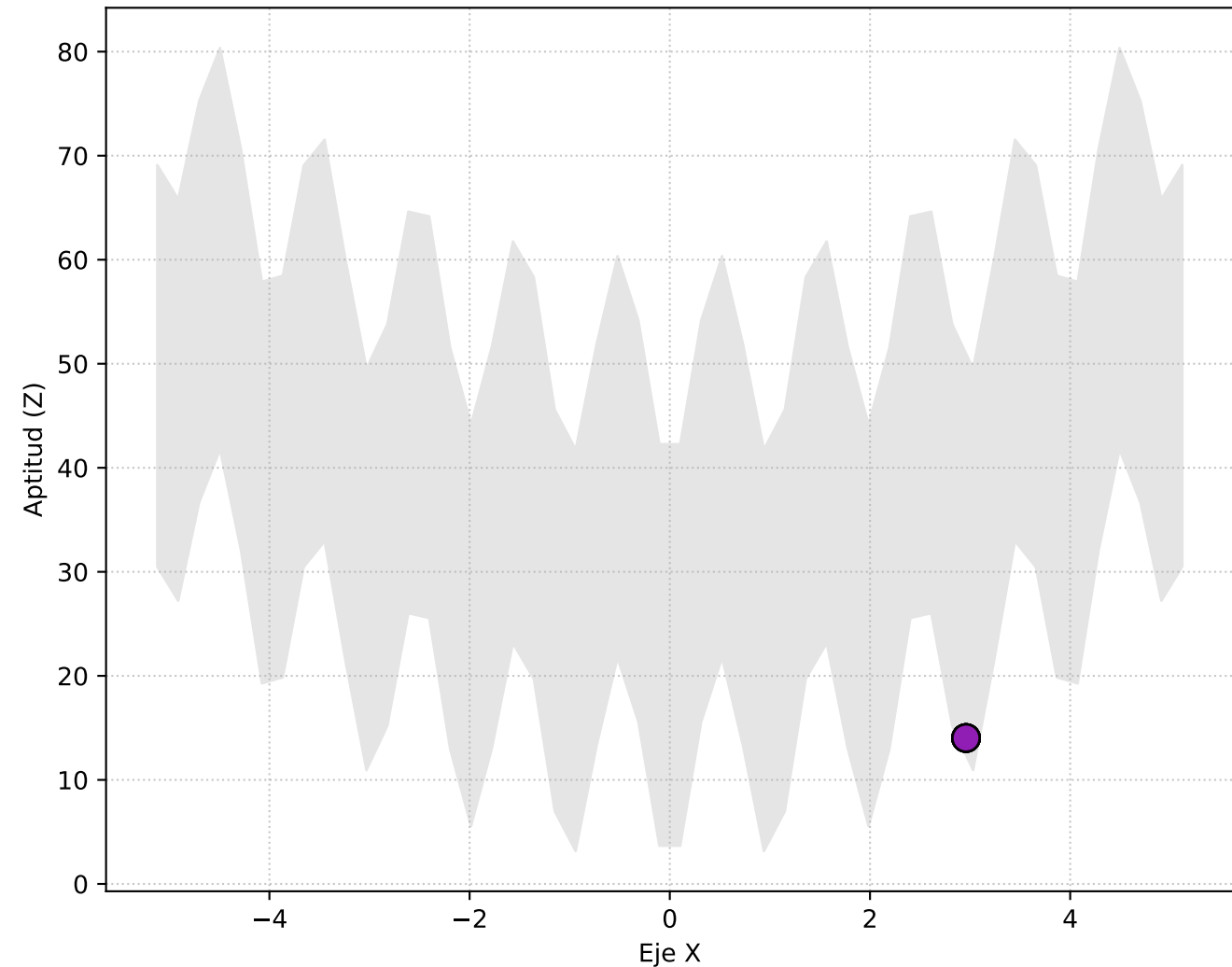
Vista Superior (X, Y)



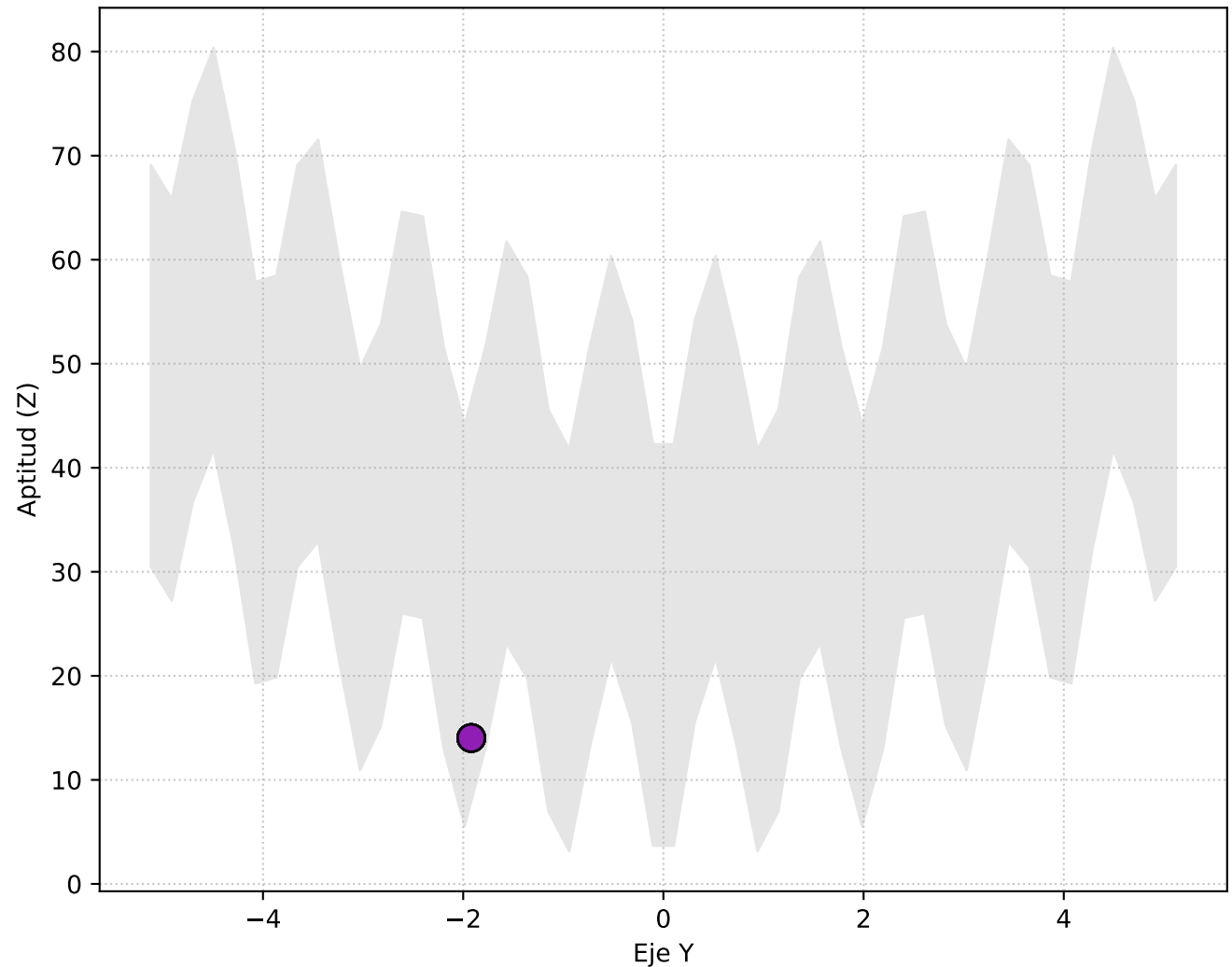
Perspectiva 3D



Vista Lateral (X vs Elevación Z)

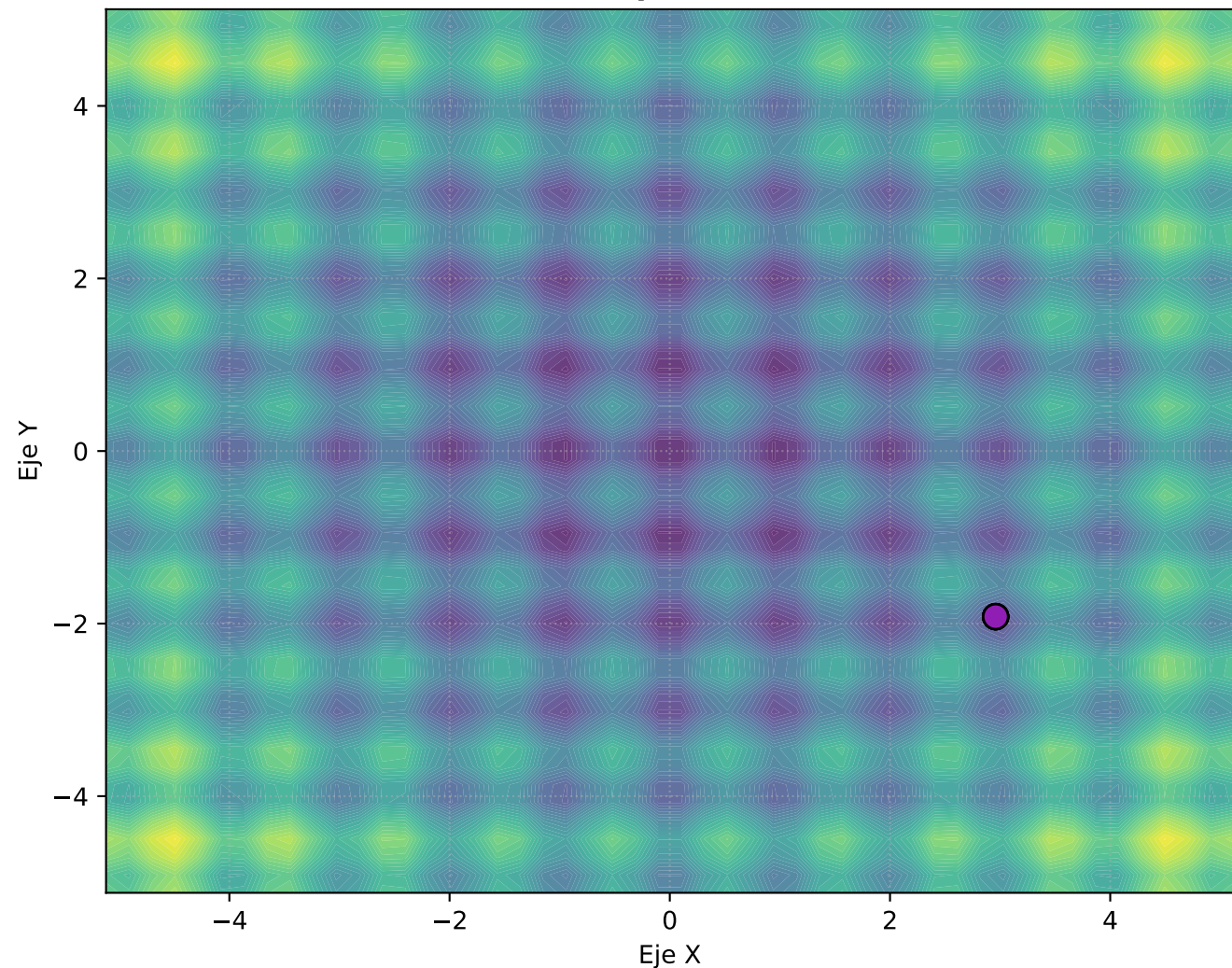


Vista Frontal (Y vs Elevación Z)

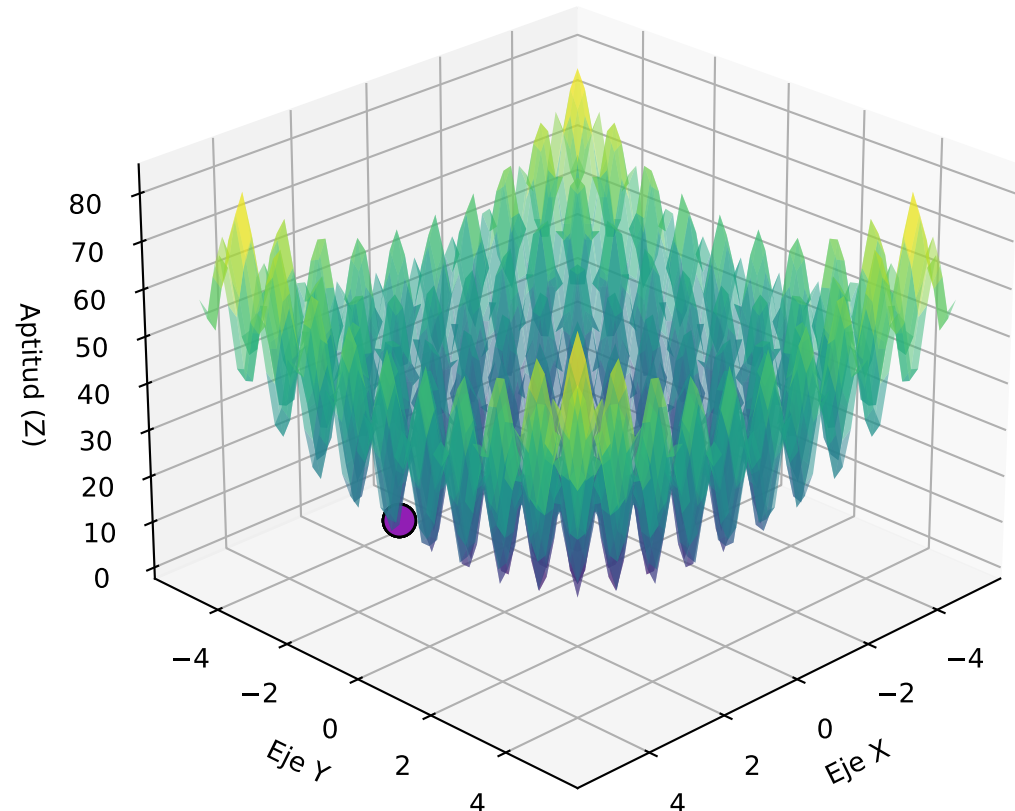


Progreso: Generación 30

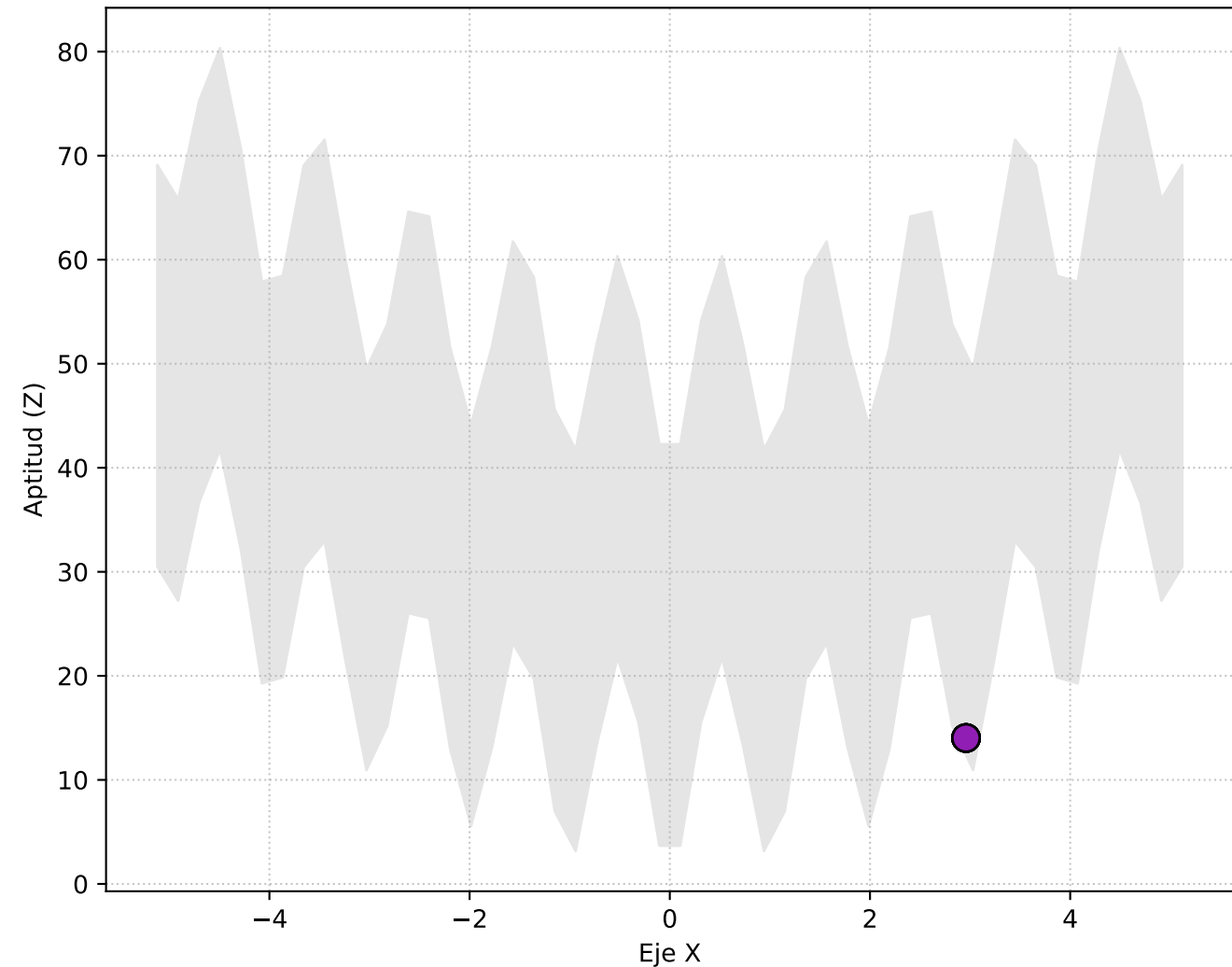
Vista Superior (X, Y)



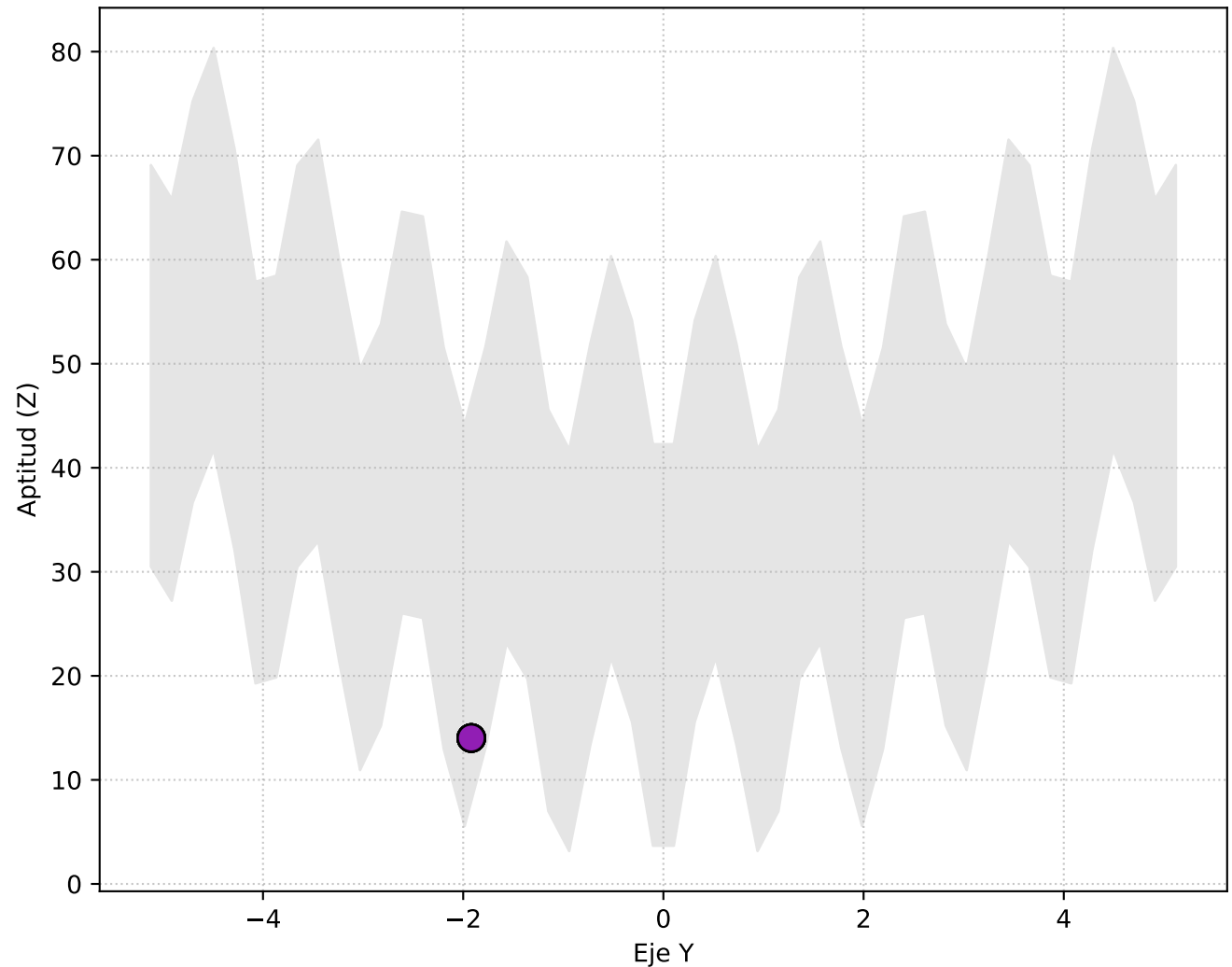
Perspectiva 3D



Vista Lateral (X vs Elevación Z)

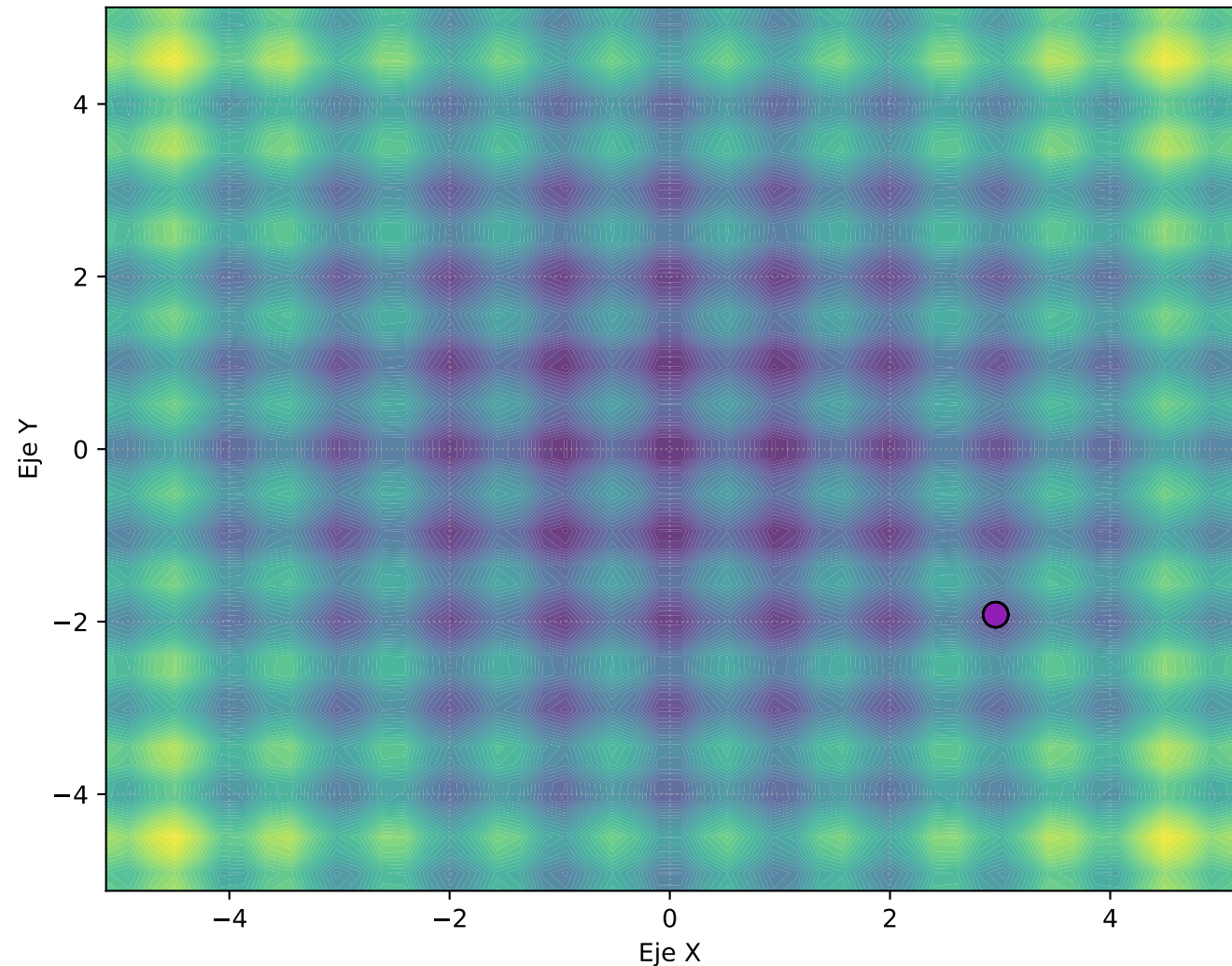


Vista Frontal (Y vs Elevación Z)

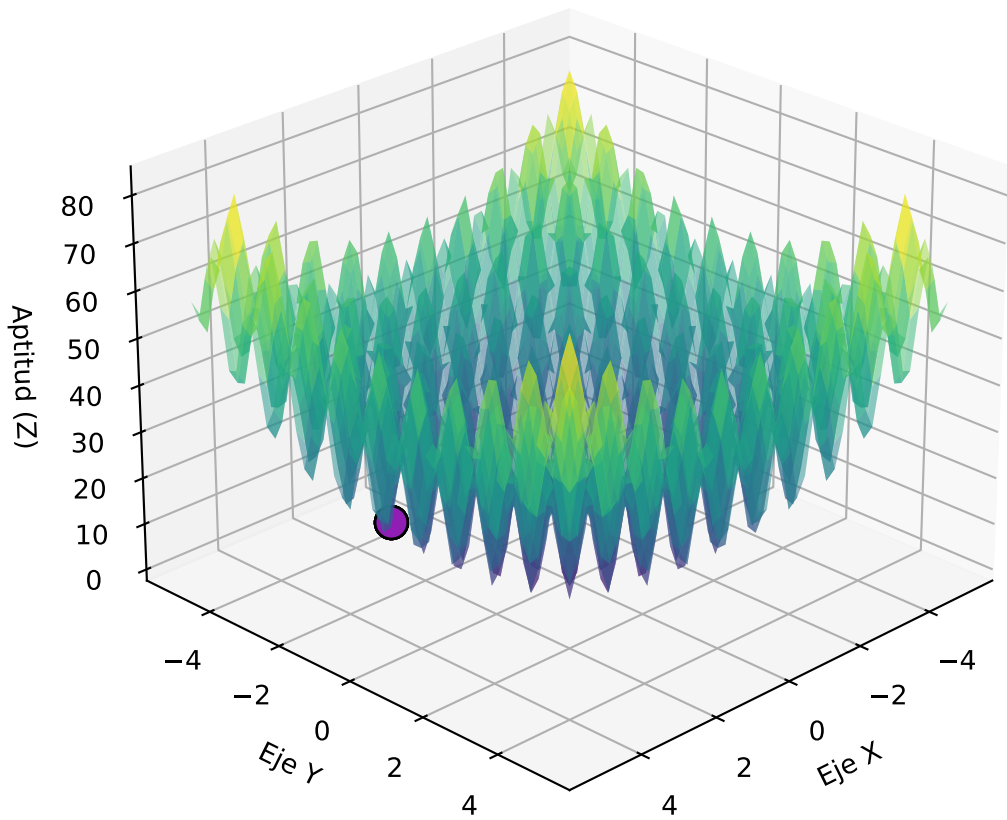


Progreso: Generación 40

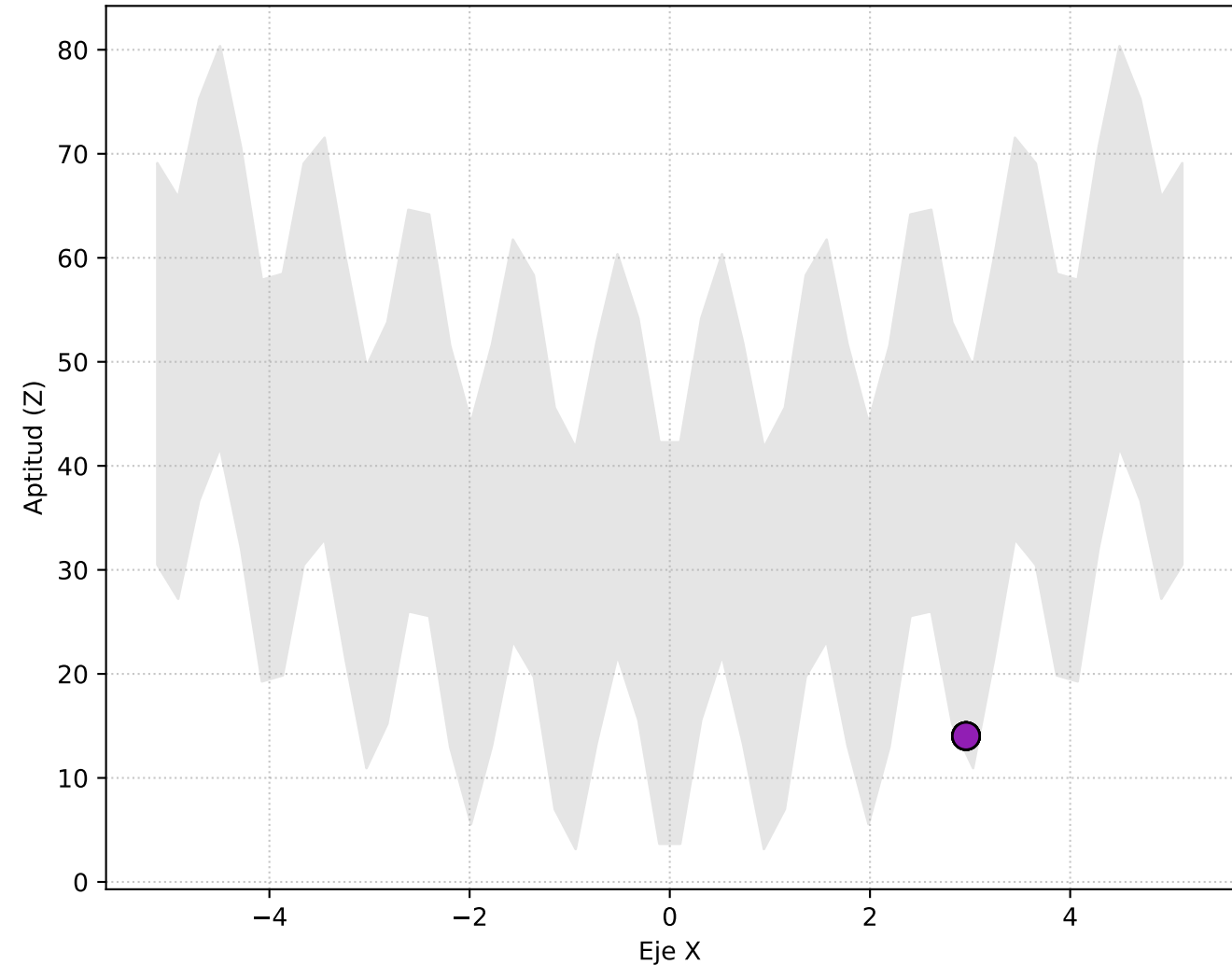
Vista Superior (X, Y)



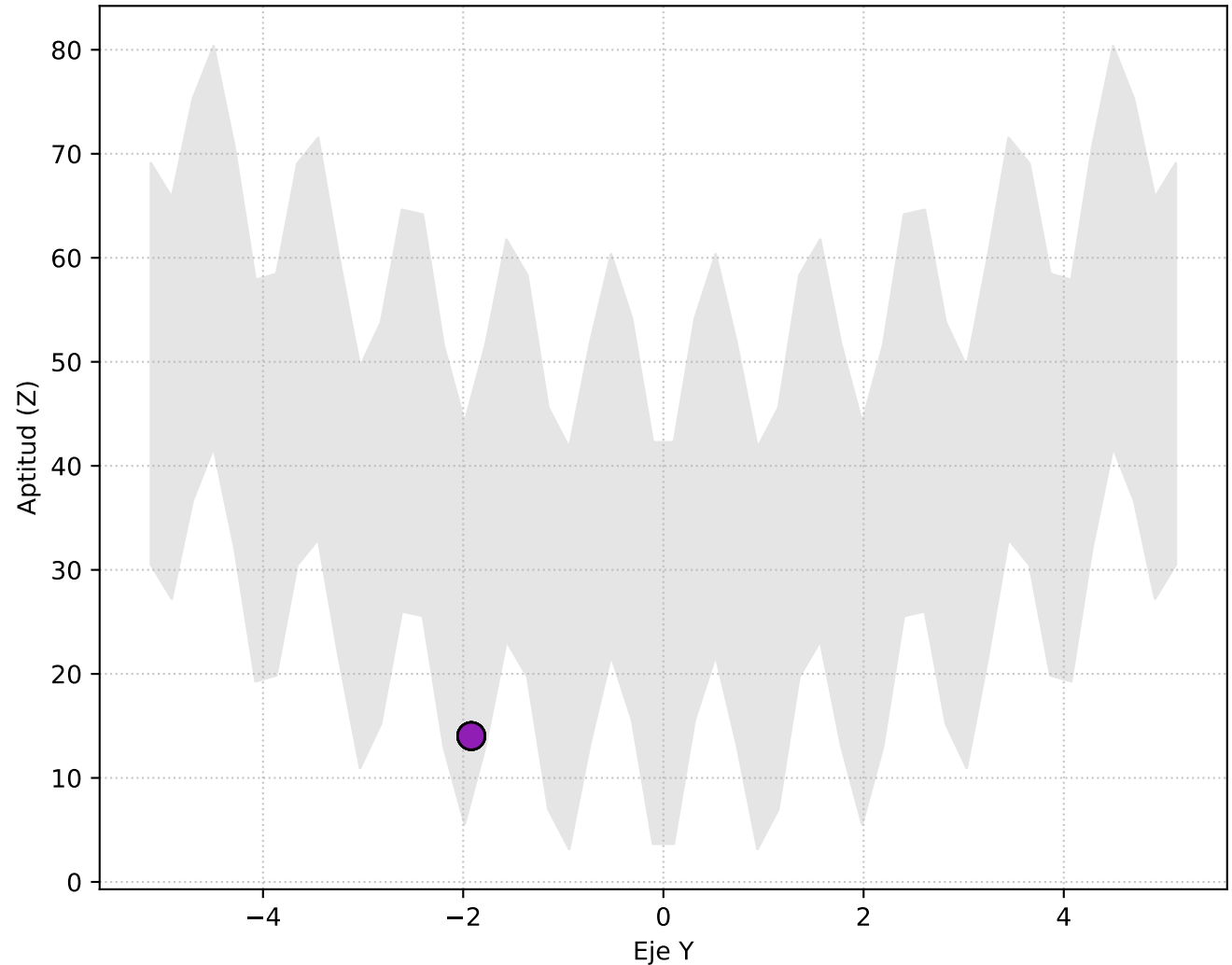
Perspectiva 3D



Vista Lateral (X vs Elevación Z)

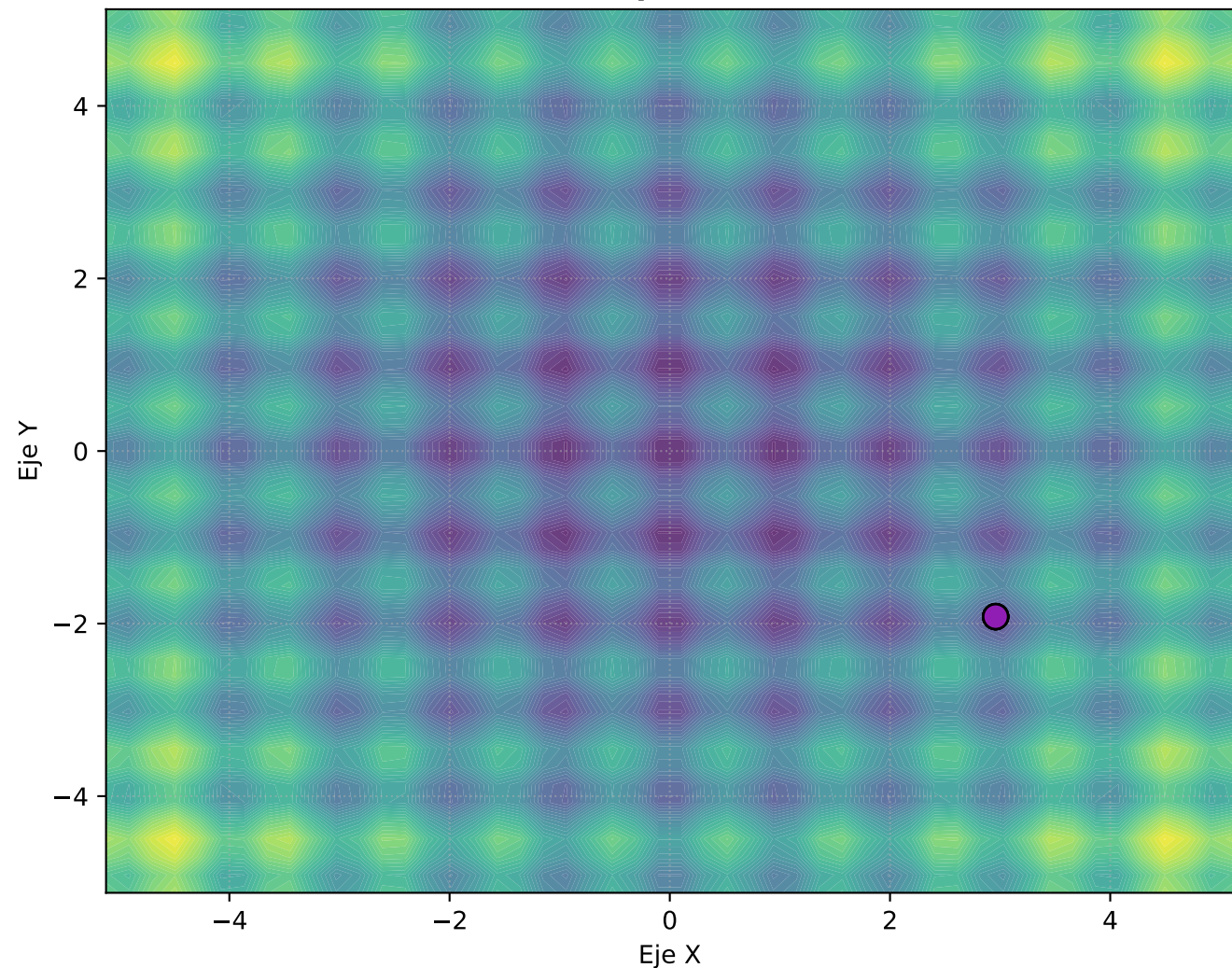


Vista Frontal (Y vs Elevación Z)

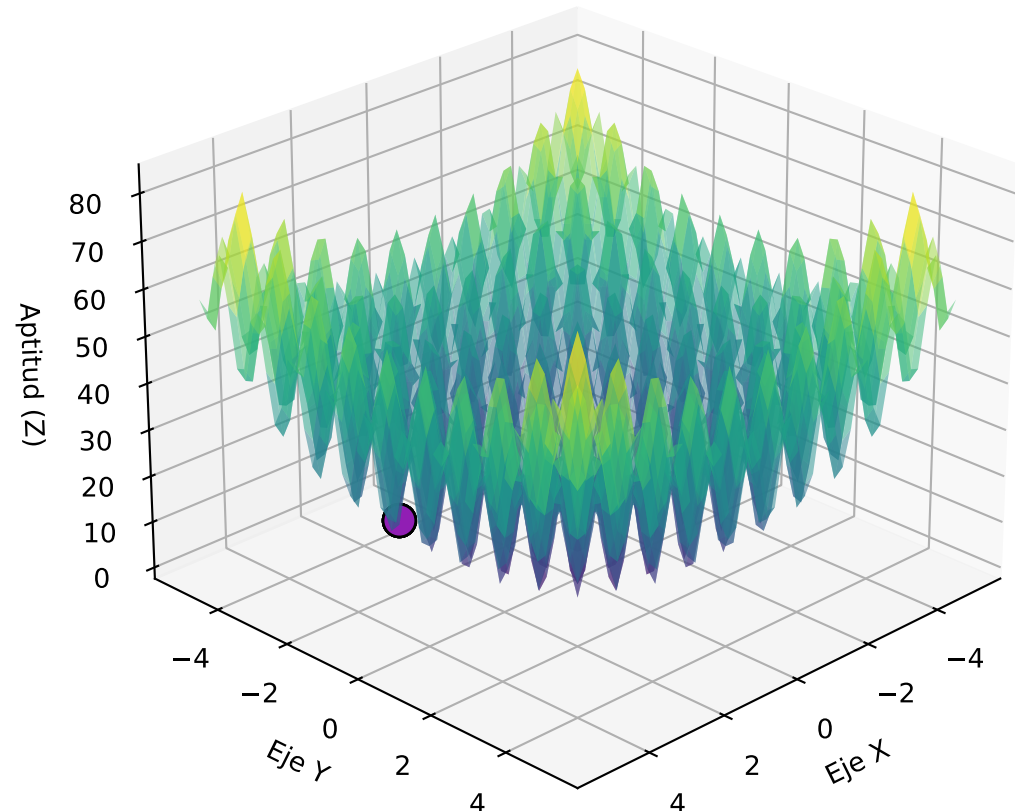


Progreso: Generación 50

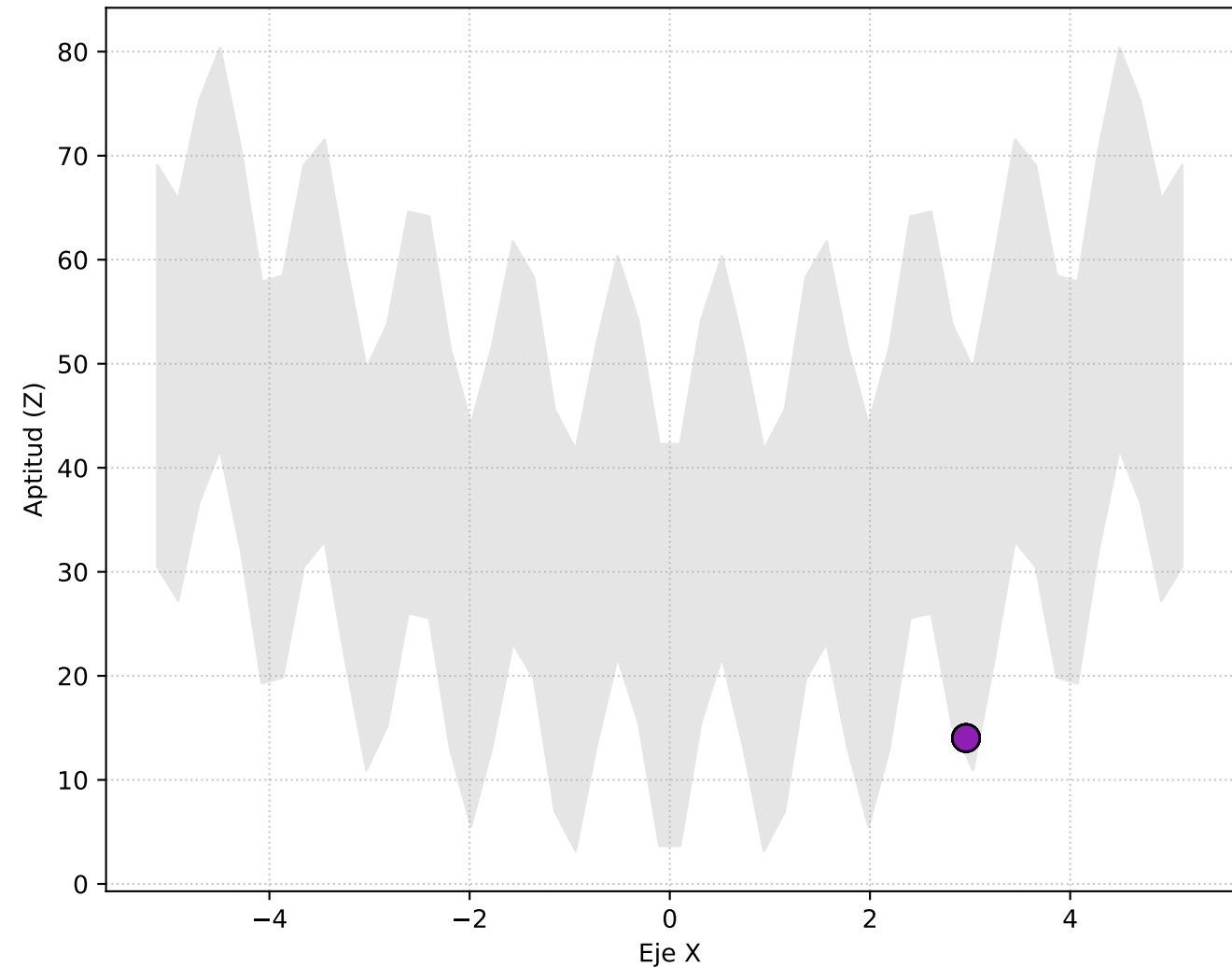
Vista Superior (X, Y)



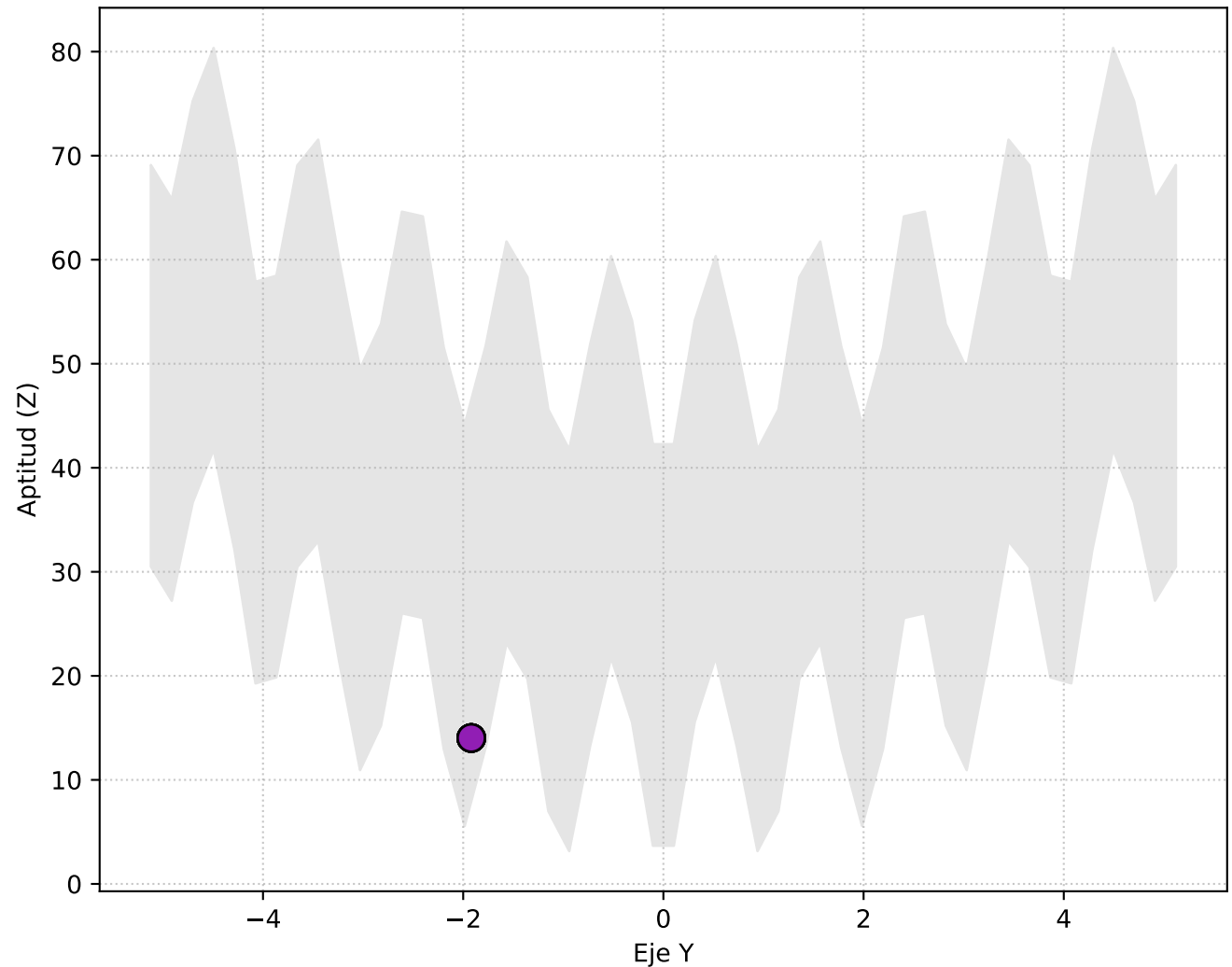
Perspectiva 3D



Vista Lateral (X vs Elevación Z)

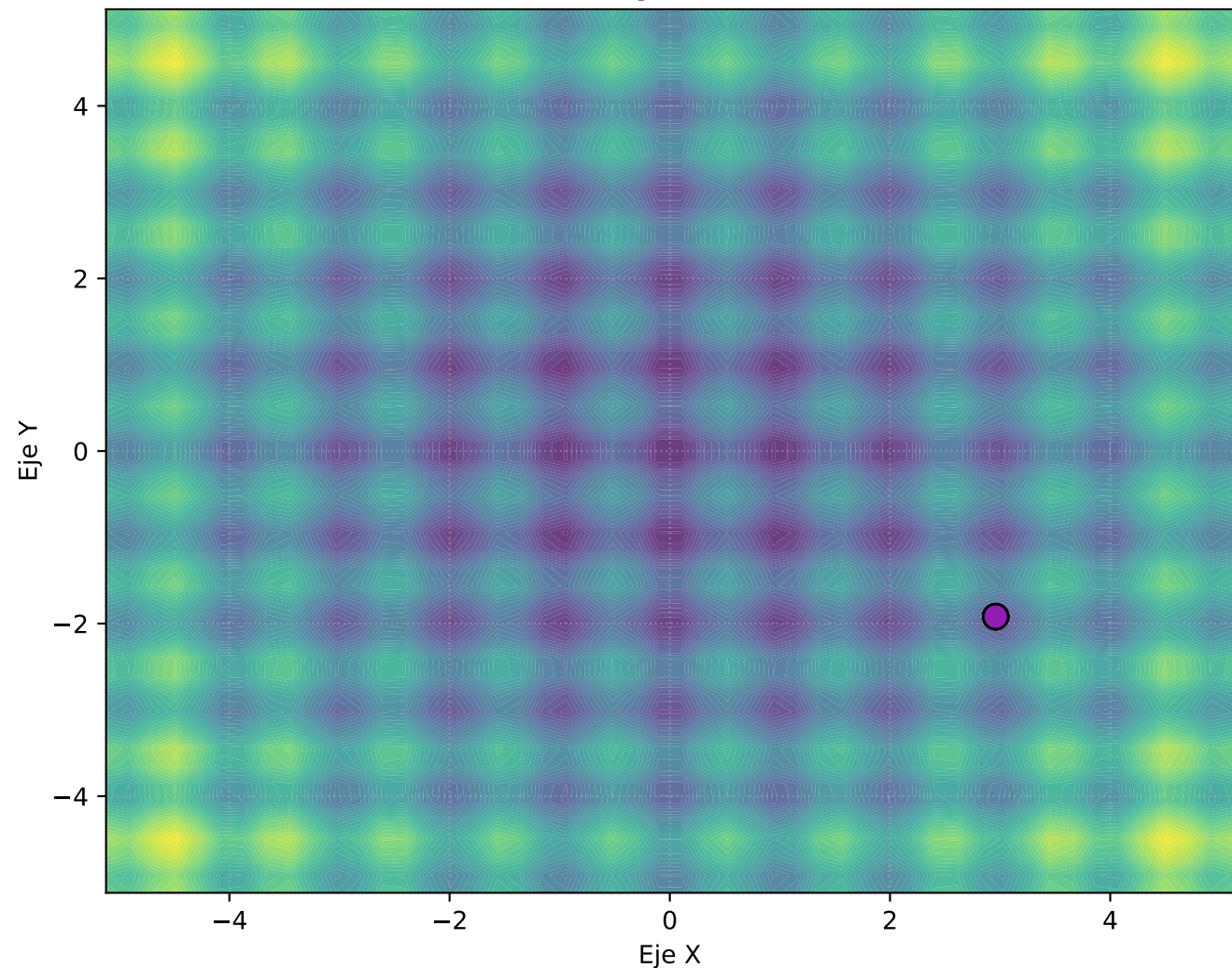


Vista Frontal (Y vs Elevación Z)

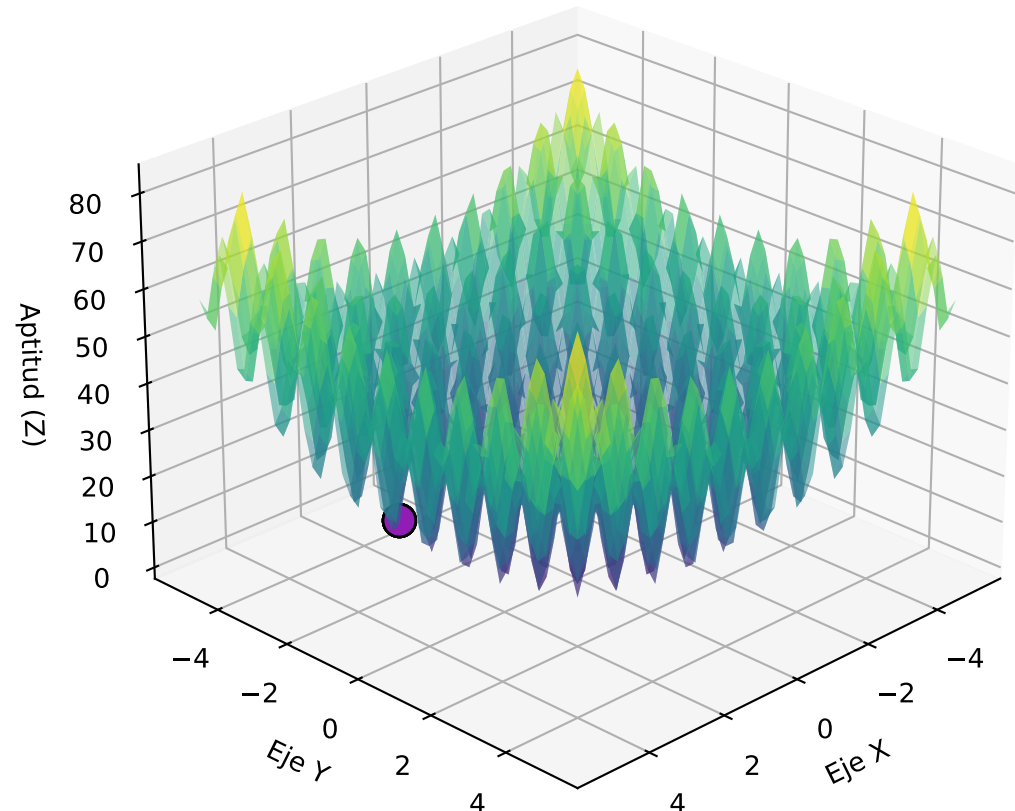


Progreso: Generación 60

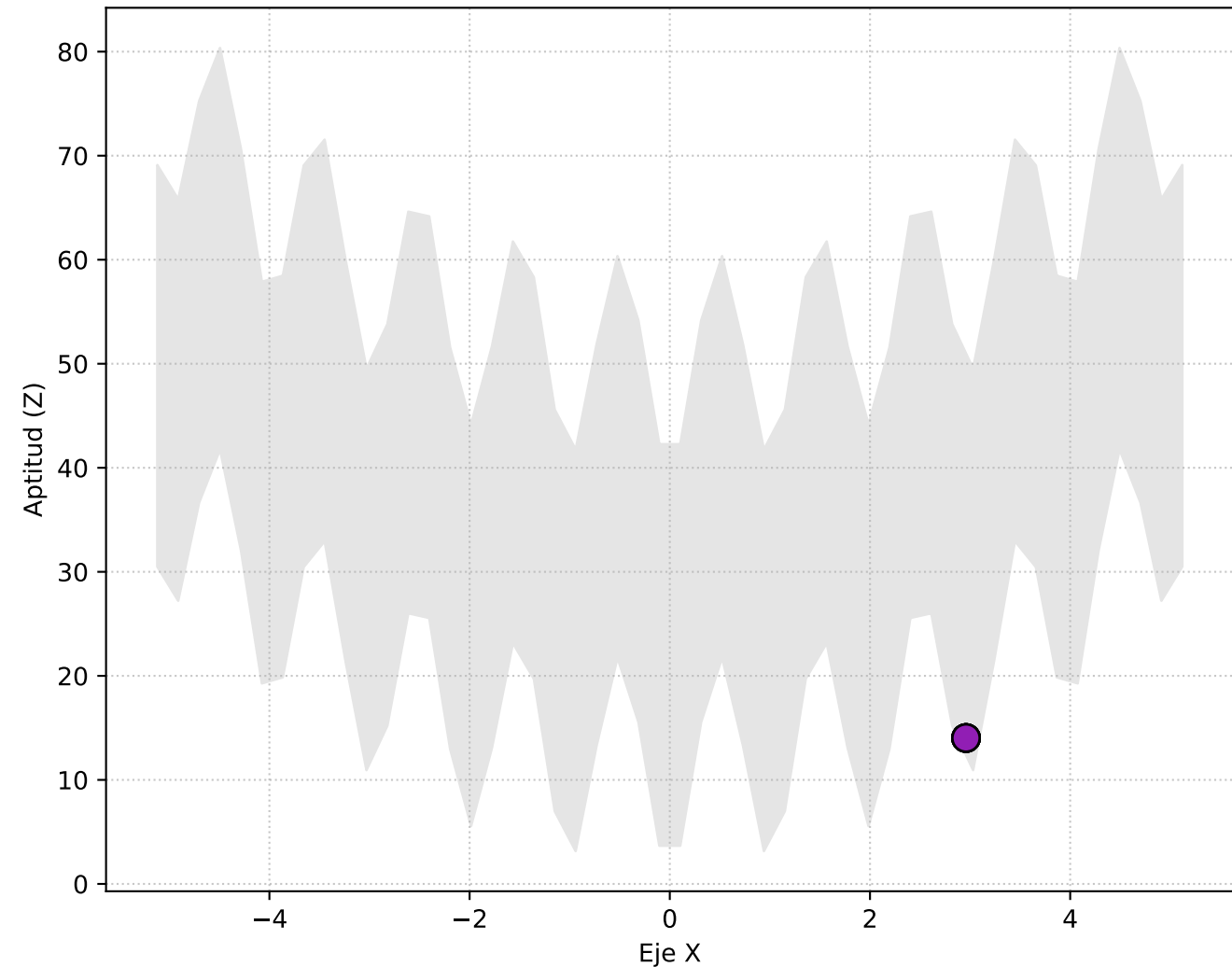
Vista Superior (X, Y)



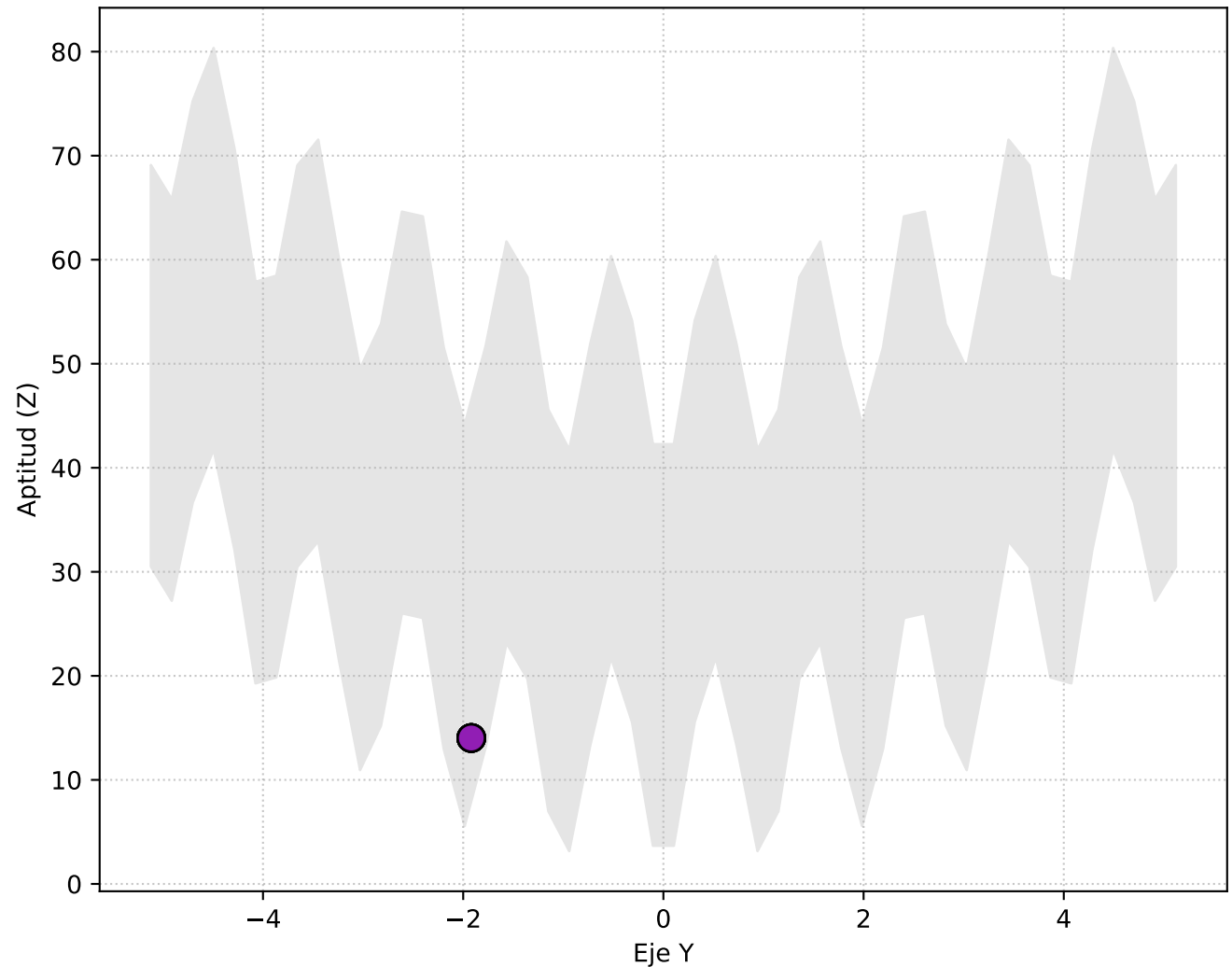
Perspectiva 3D



Vista Lateral (X vs Elevación Z)

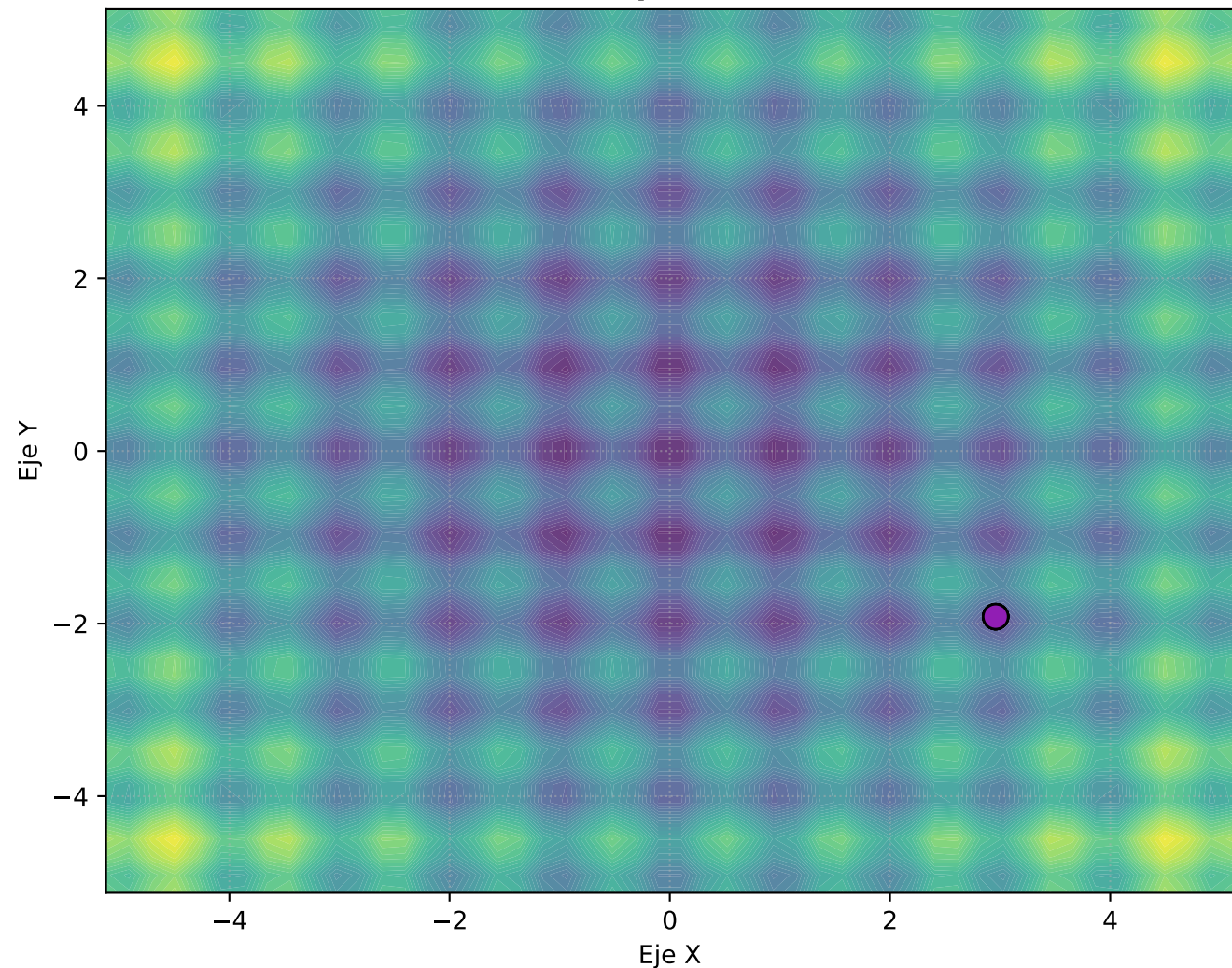


Vista Frontal (Y vs Elevación Z)

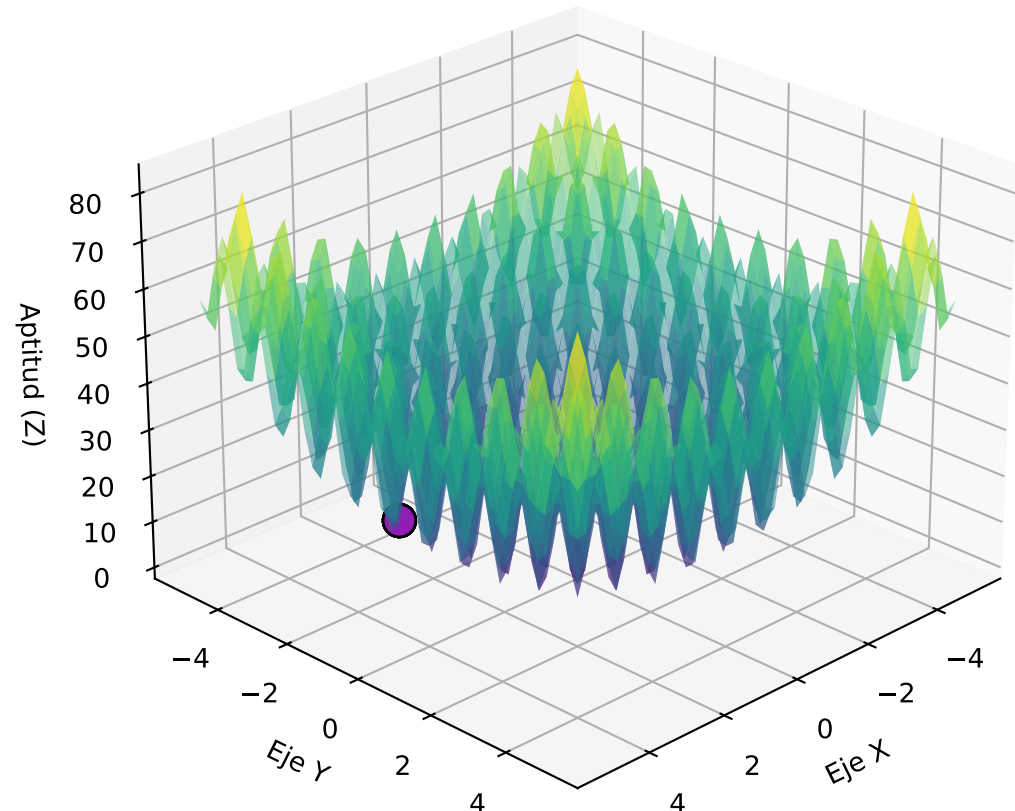


Progreso: Generación 70

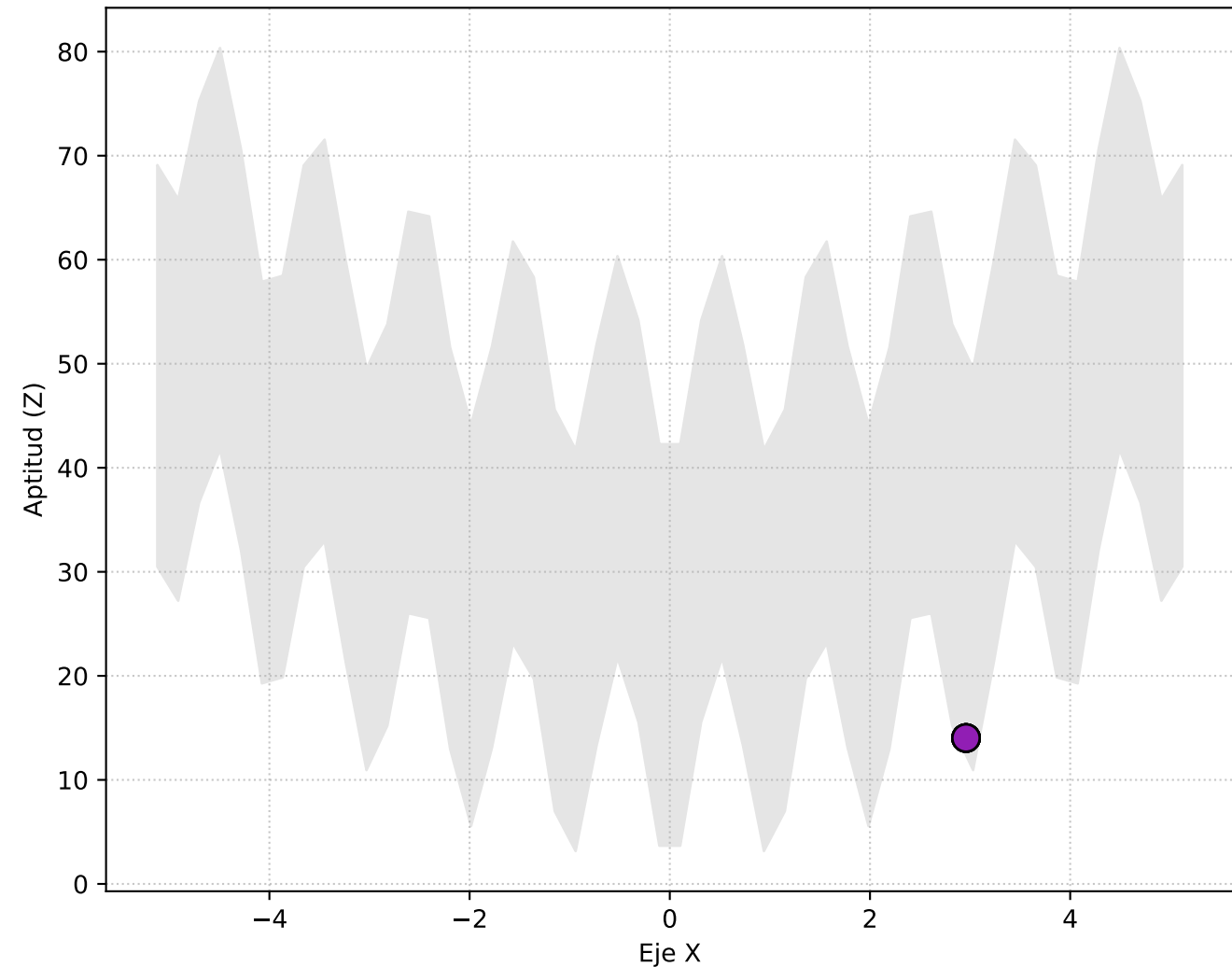
Vista Superior (X, Y)



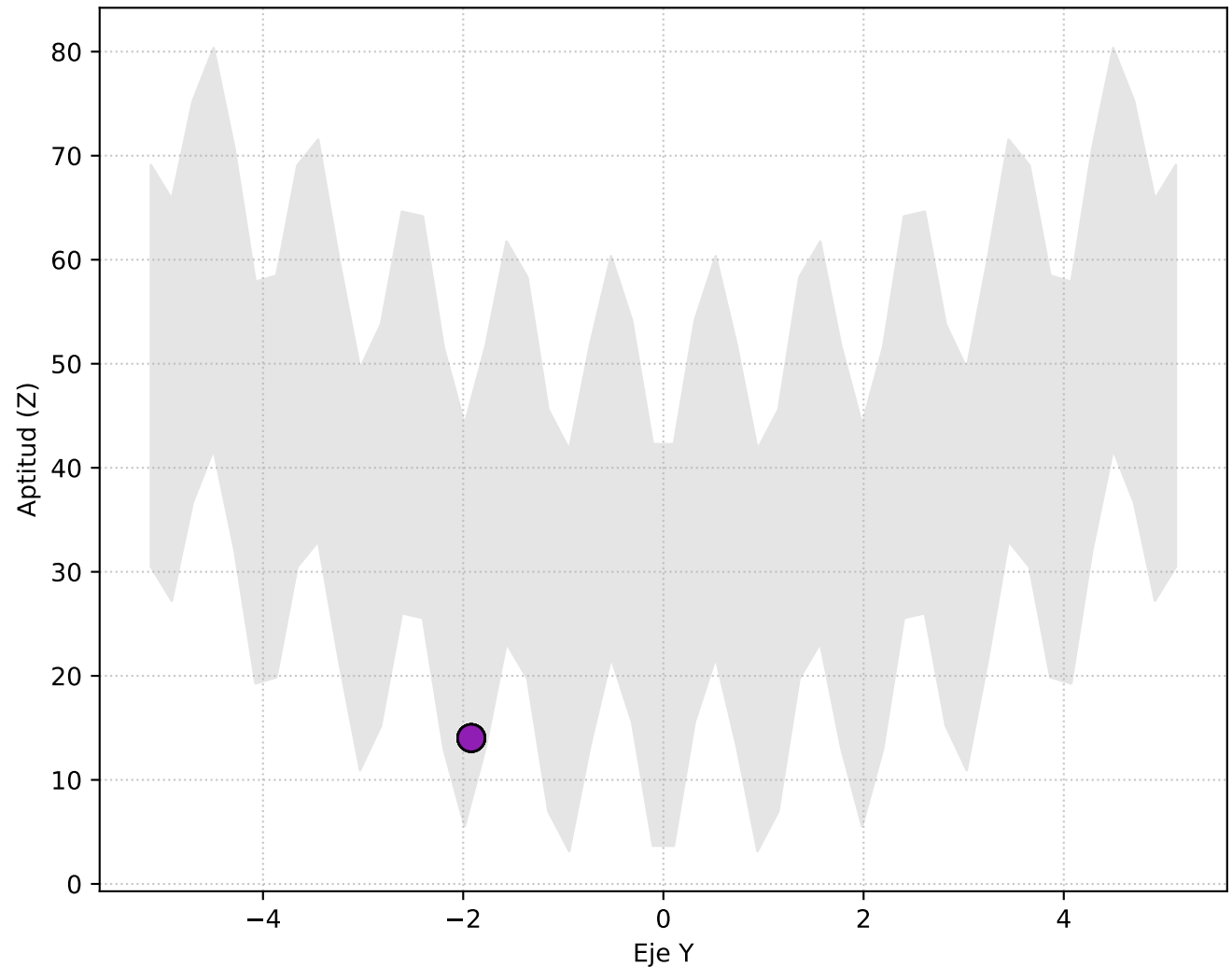
Perspectiva 3D



Vista Lateral (X vs Elevación Z)

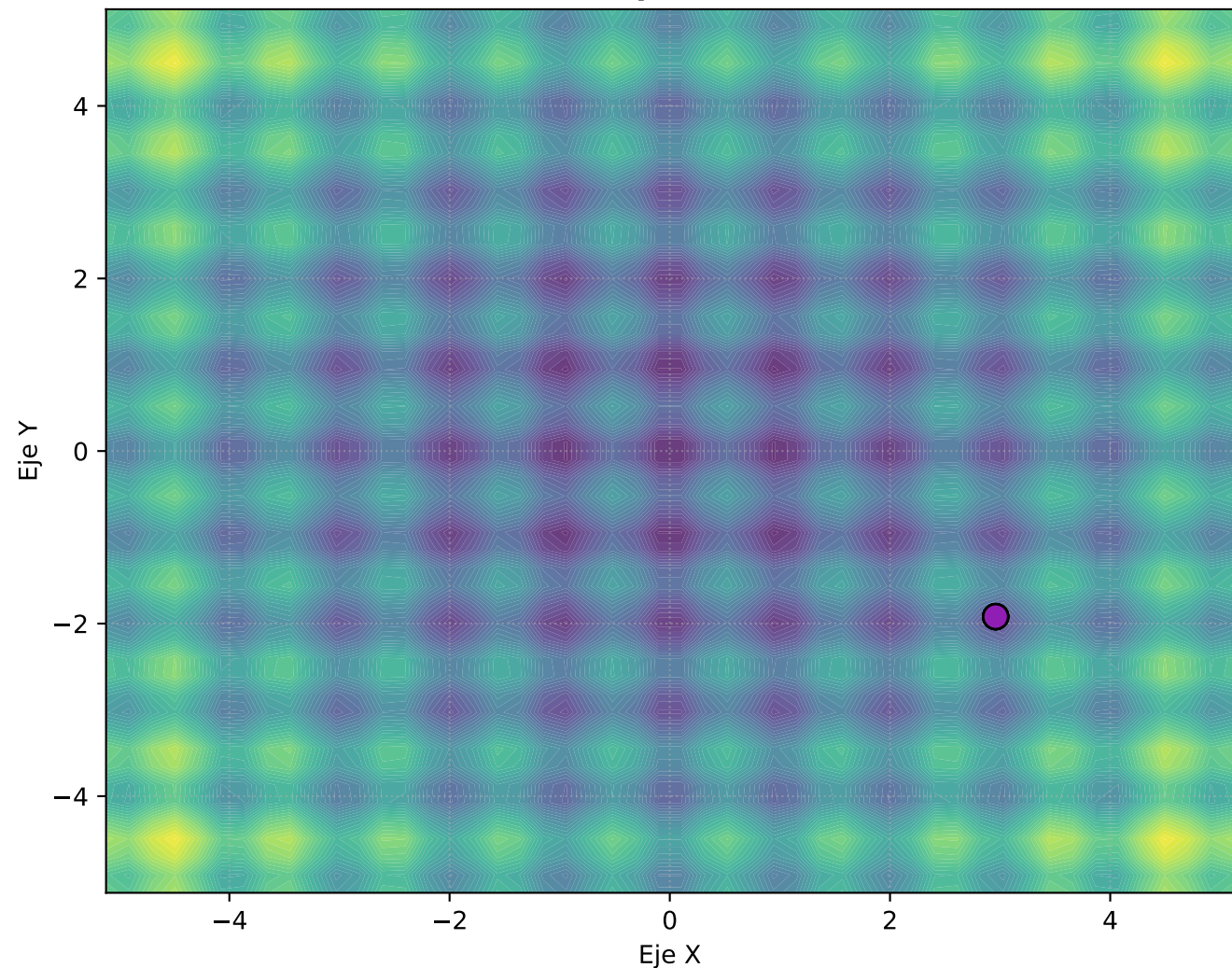


Vista Frontal (Y vs Elevación Z)

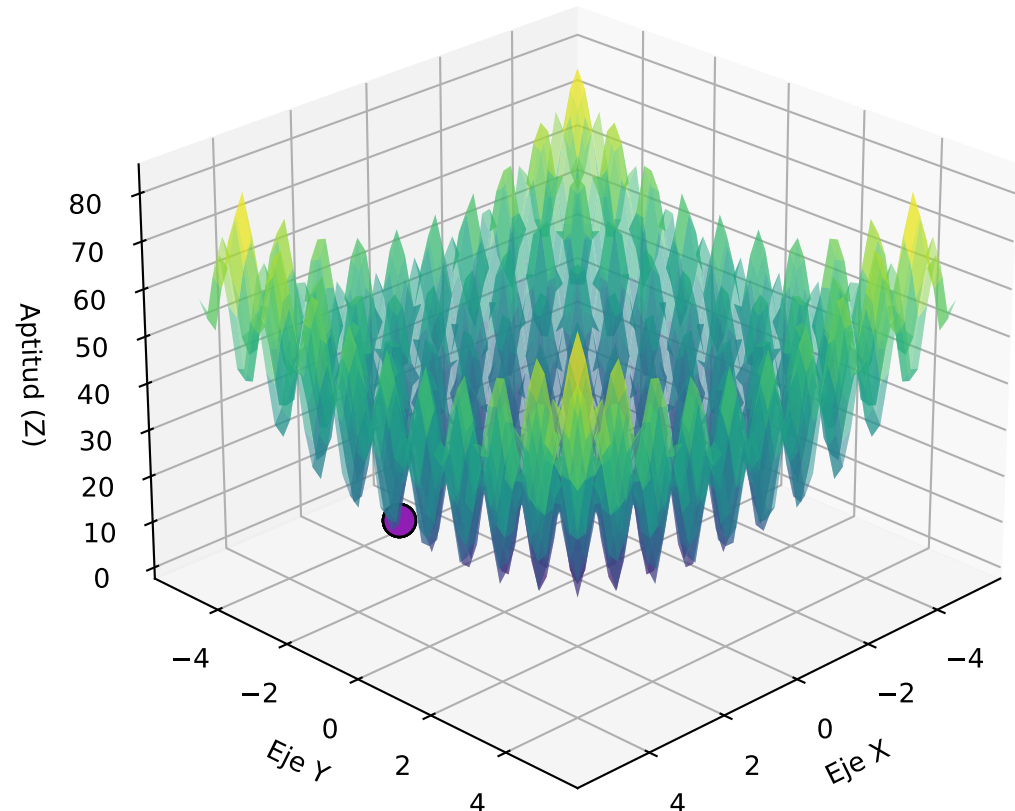


Progreso: Generación 80

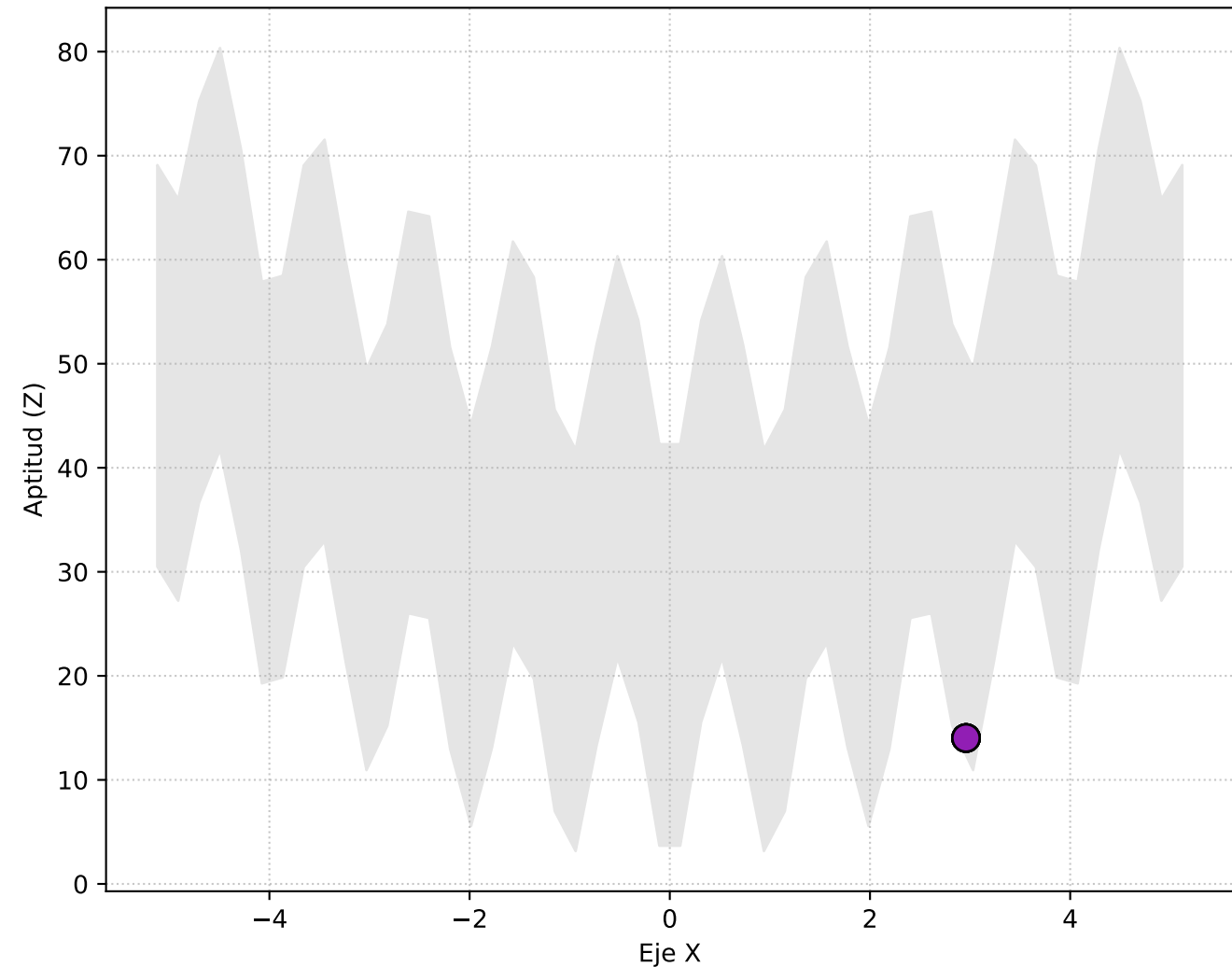
Vista Superior (X, Y)



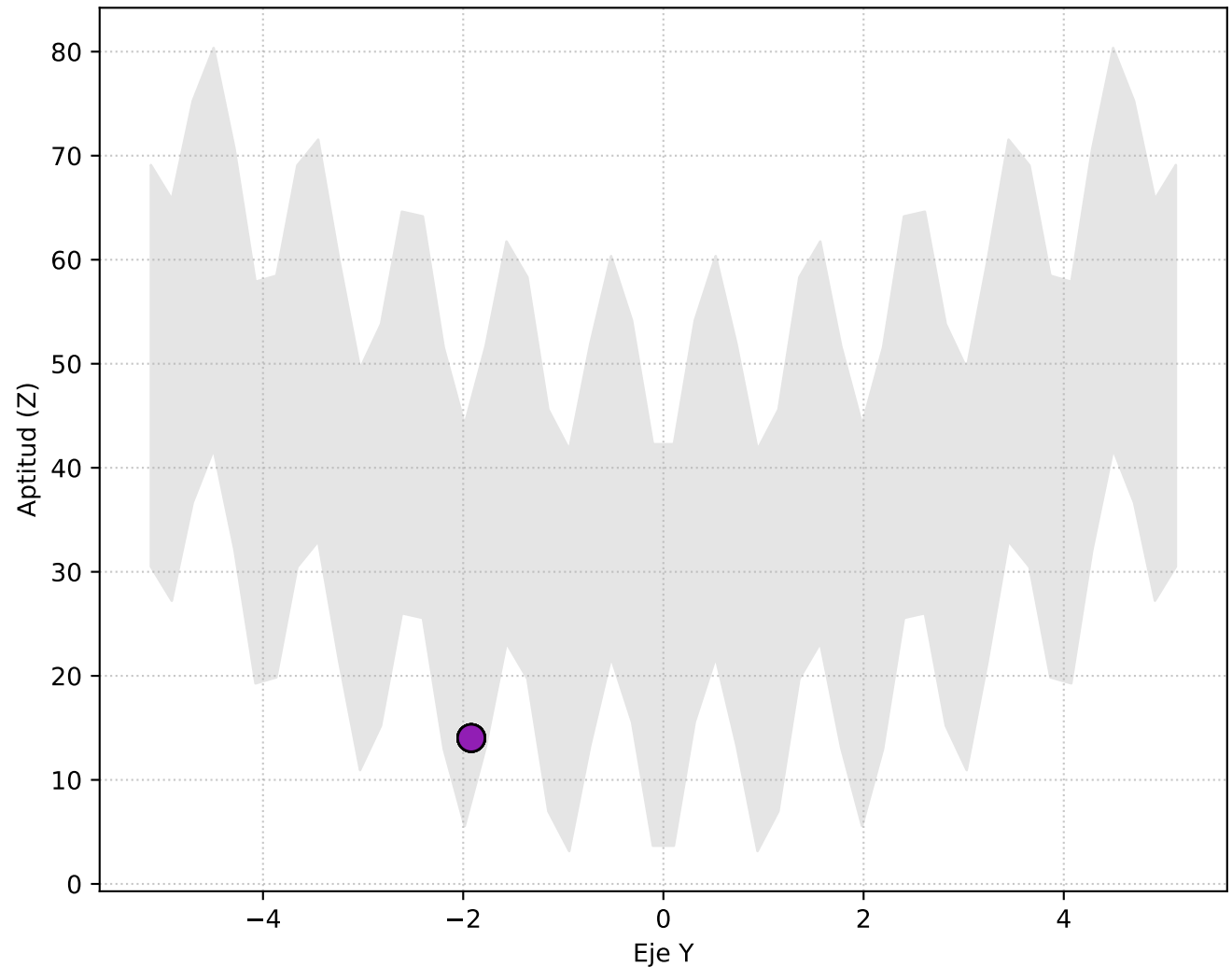
Perspectiva 3D



Vista Lateral (X vs Elevación Z)

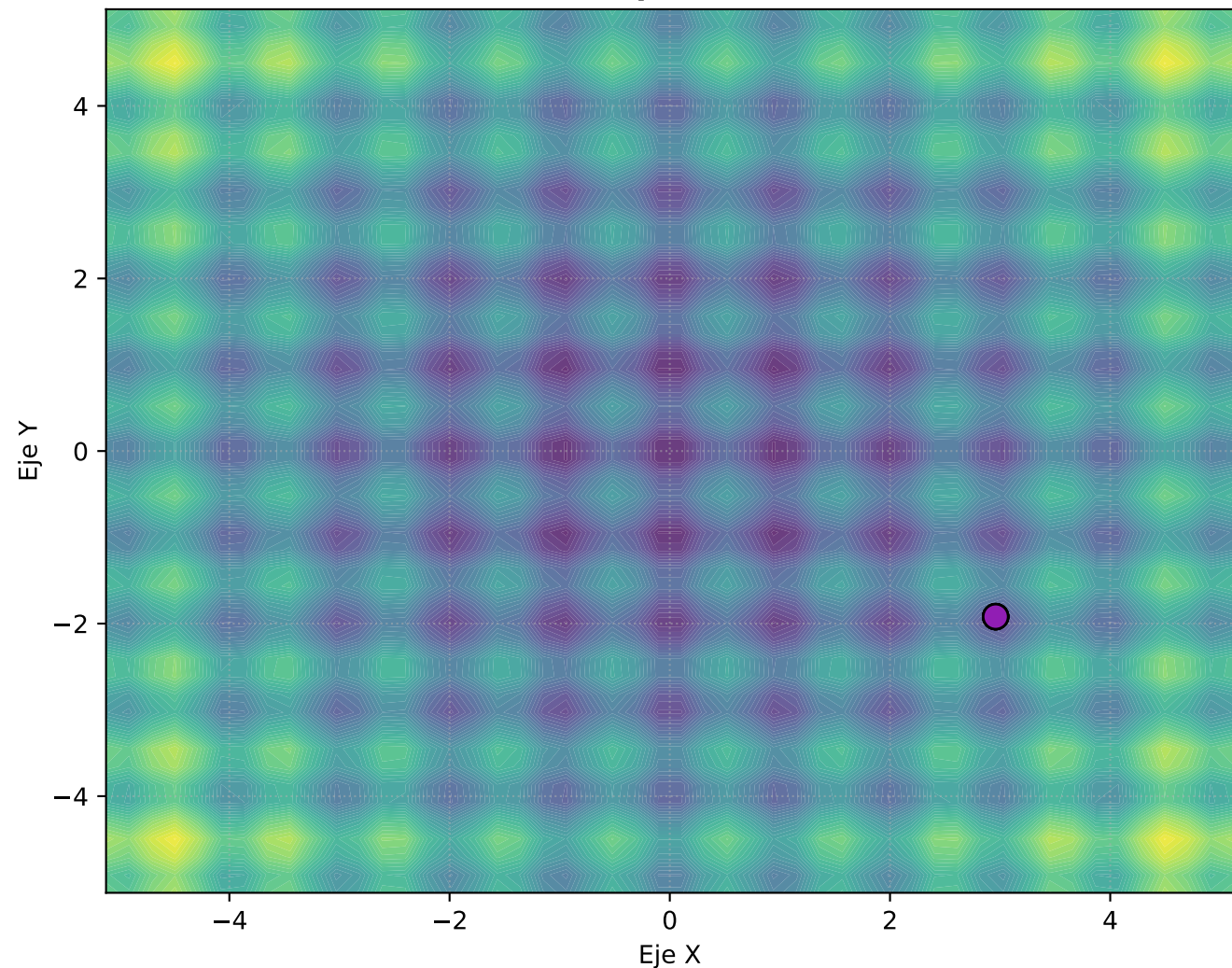


Vista Frontal (Y vs Elevación Z)

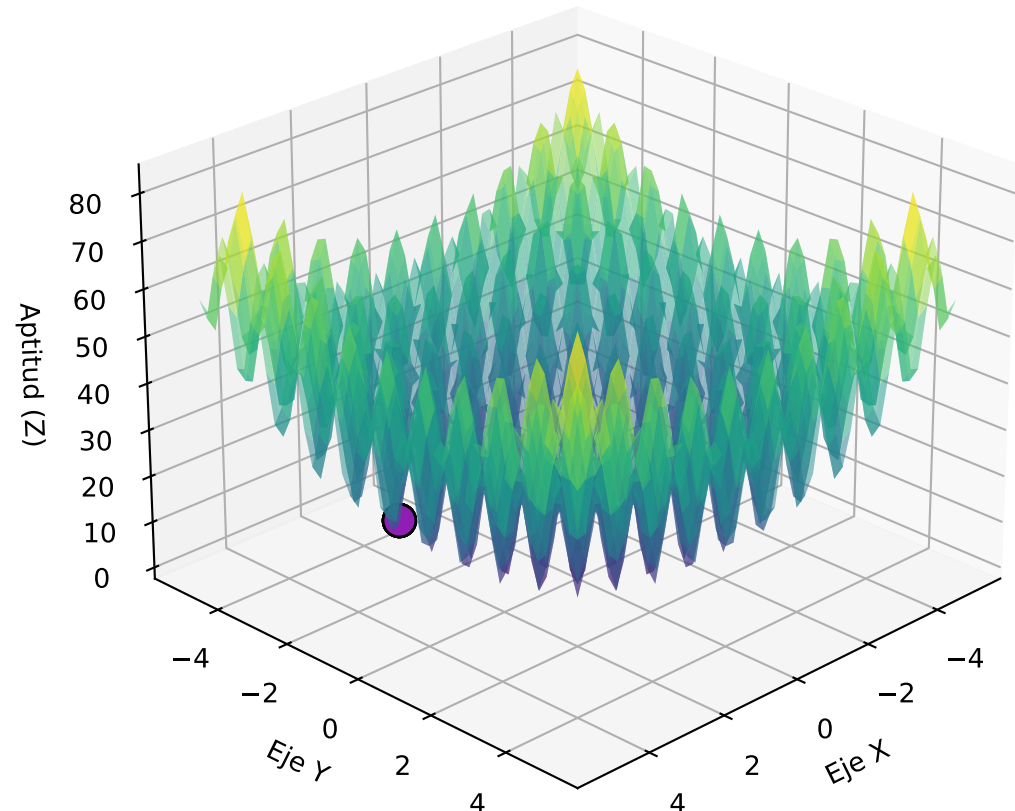


Progreso: Generación 90

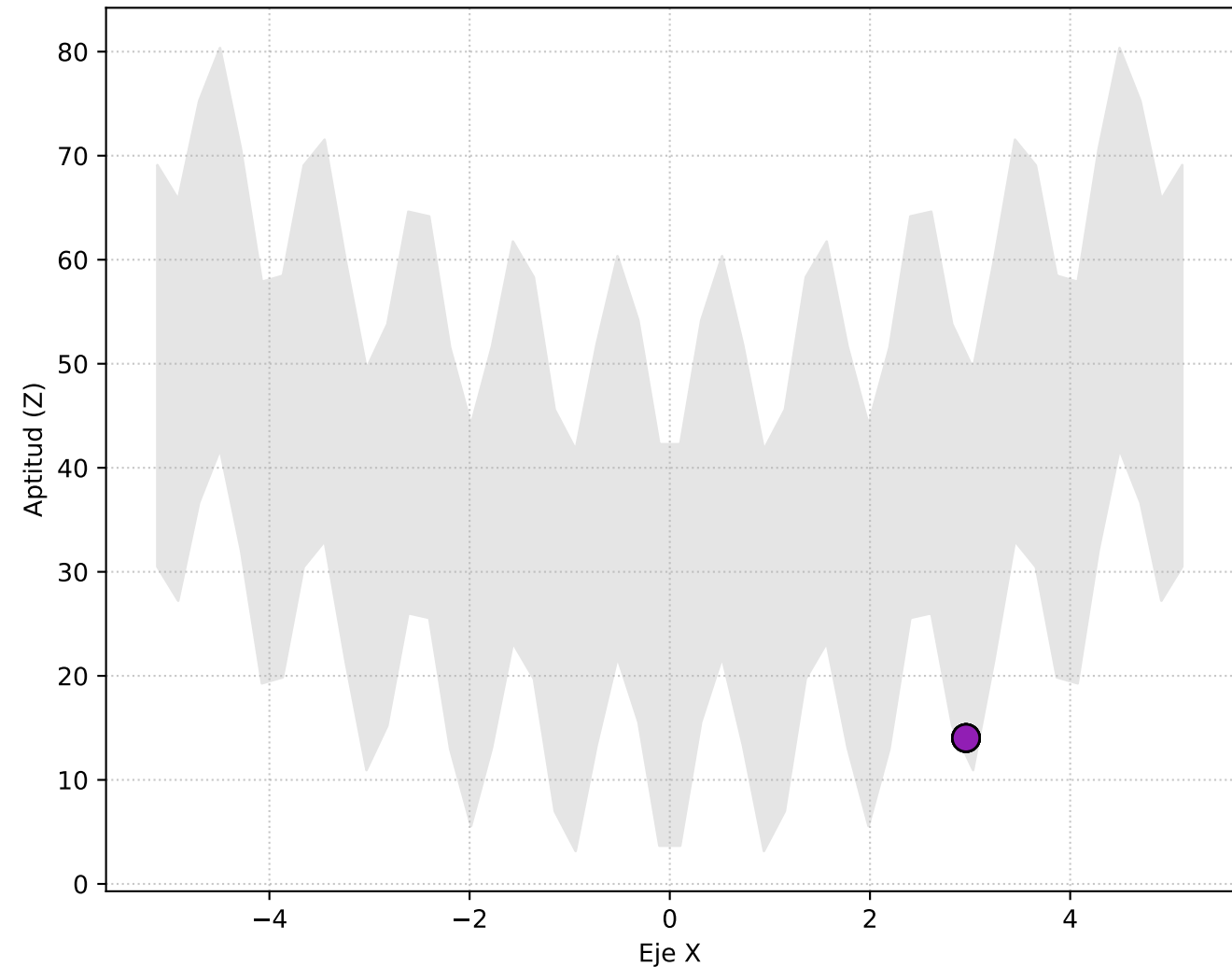
Vista Superior (X, Y)



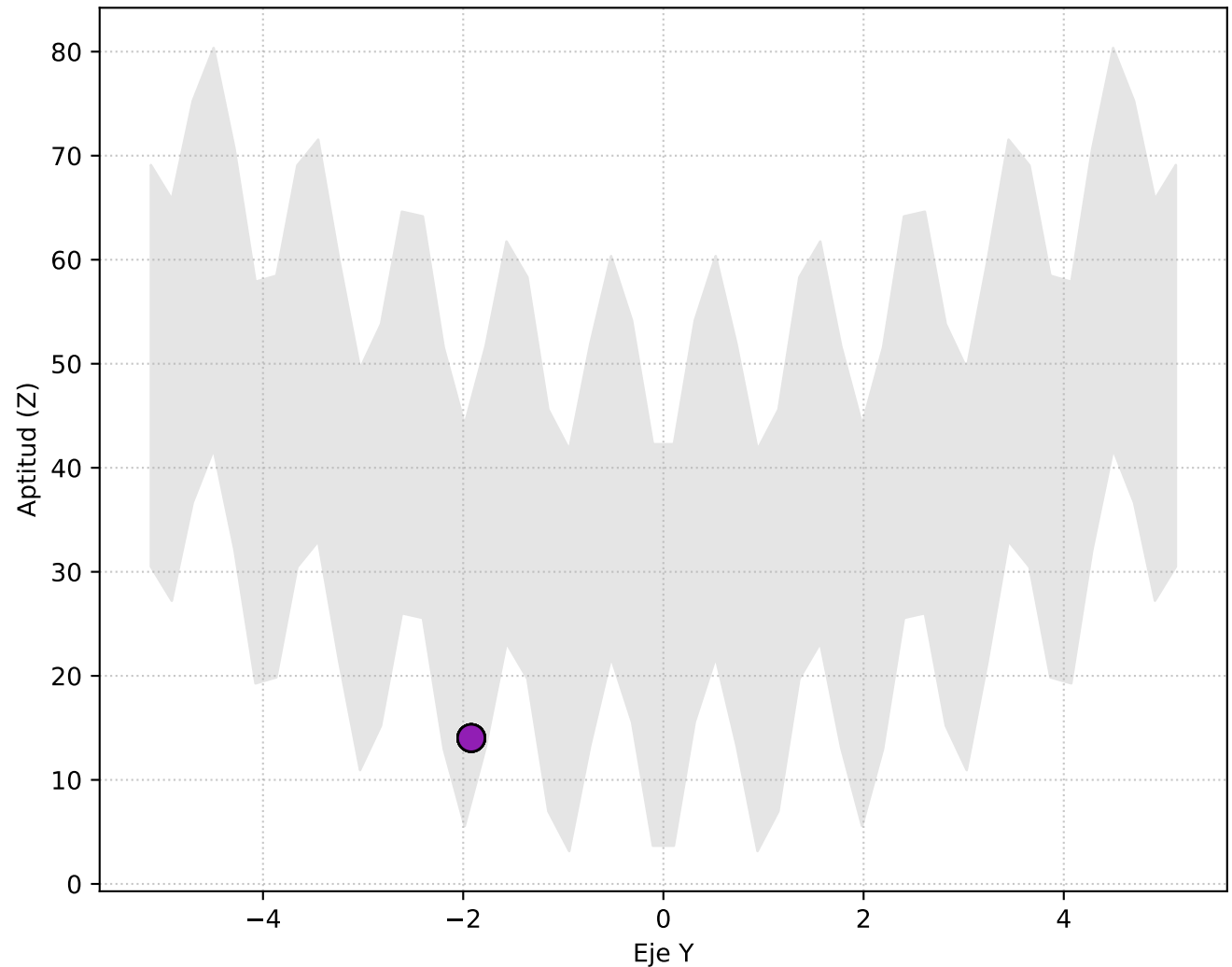
Perspectiva 3D



Vista Lateral (X vs Elevación Z)

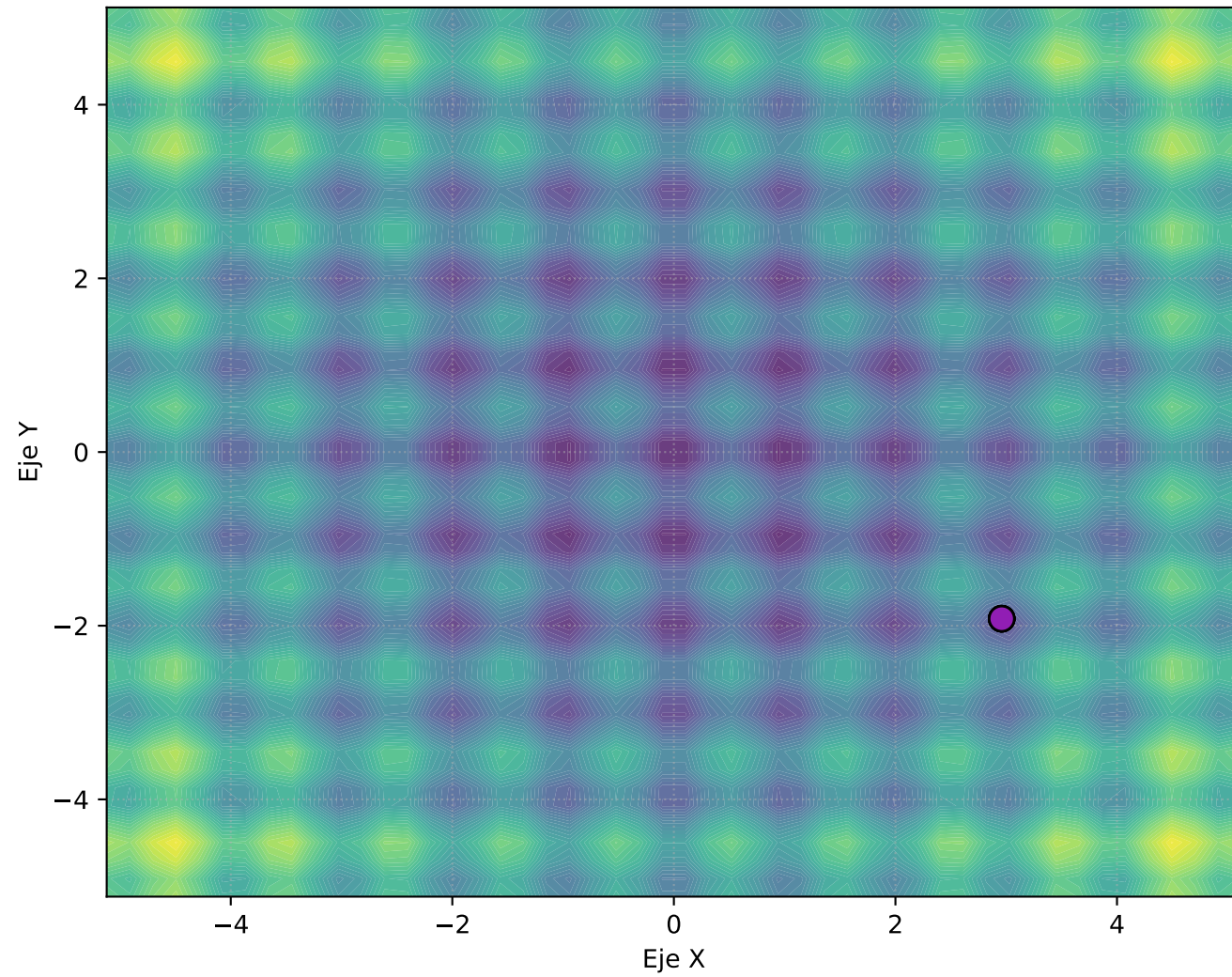


Vista Frontal (Y vs Elevación Z)

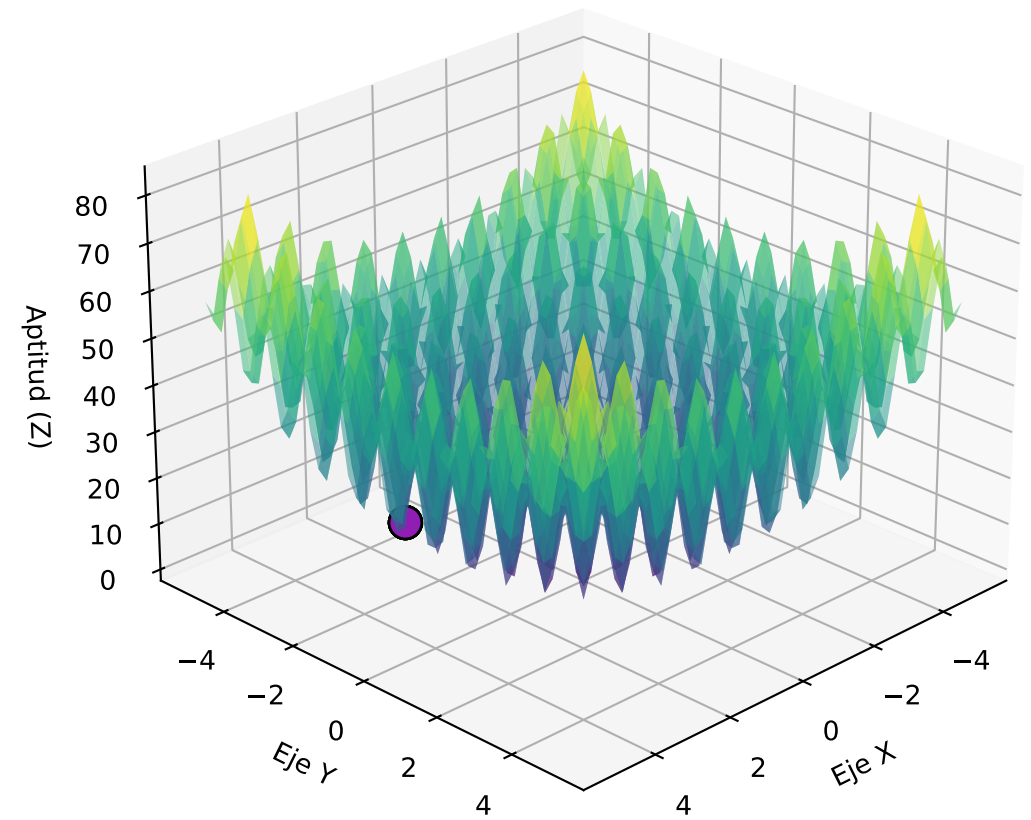


Progreso: Generación 100

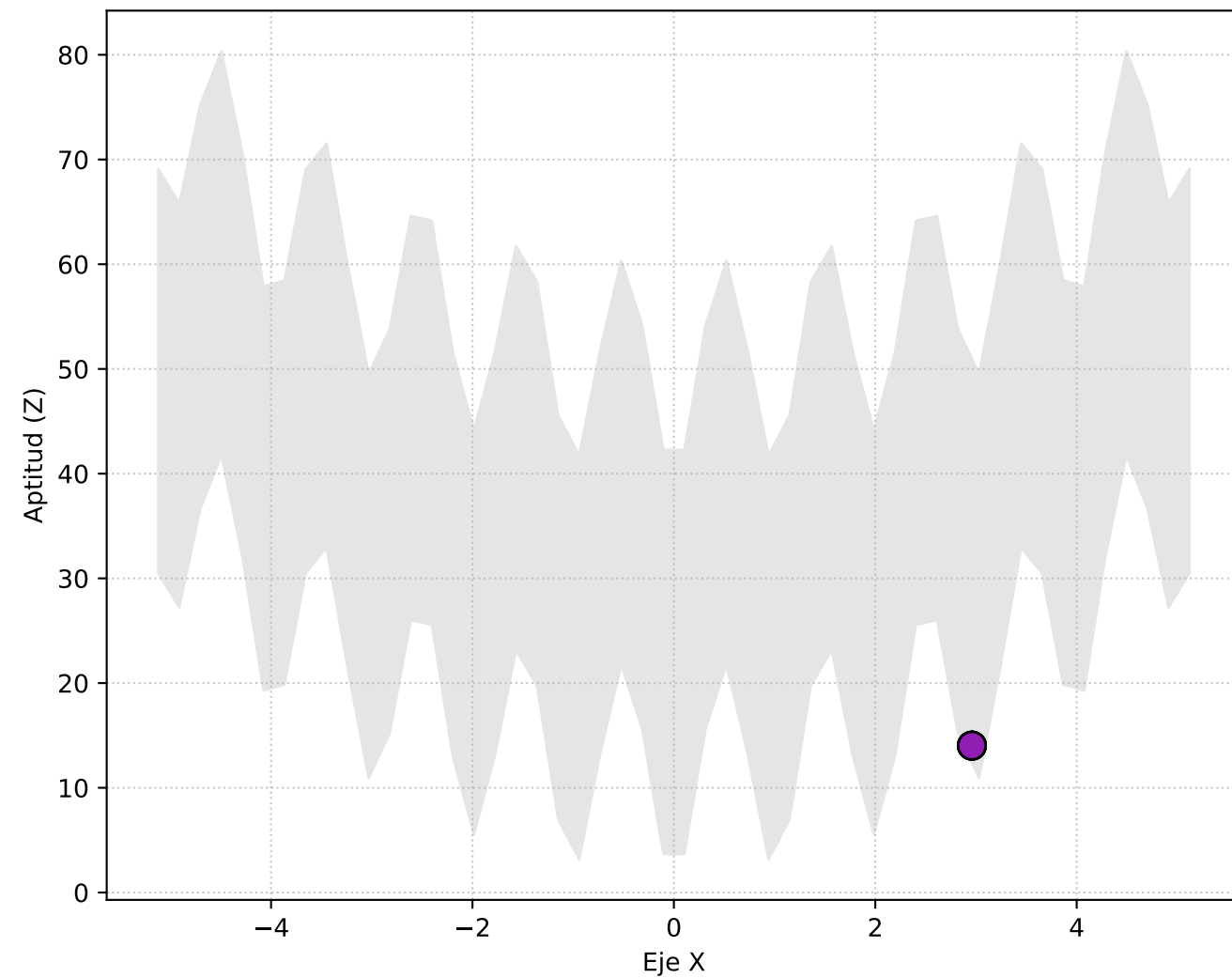
Vista Superior (X, Y)



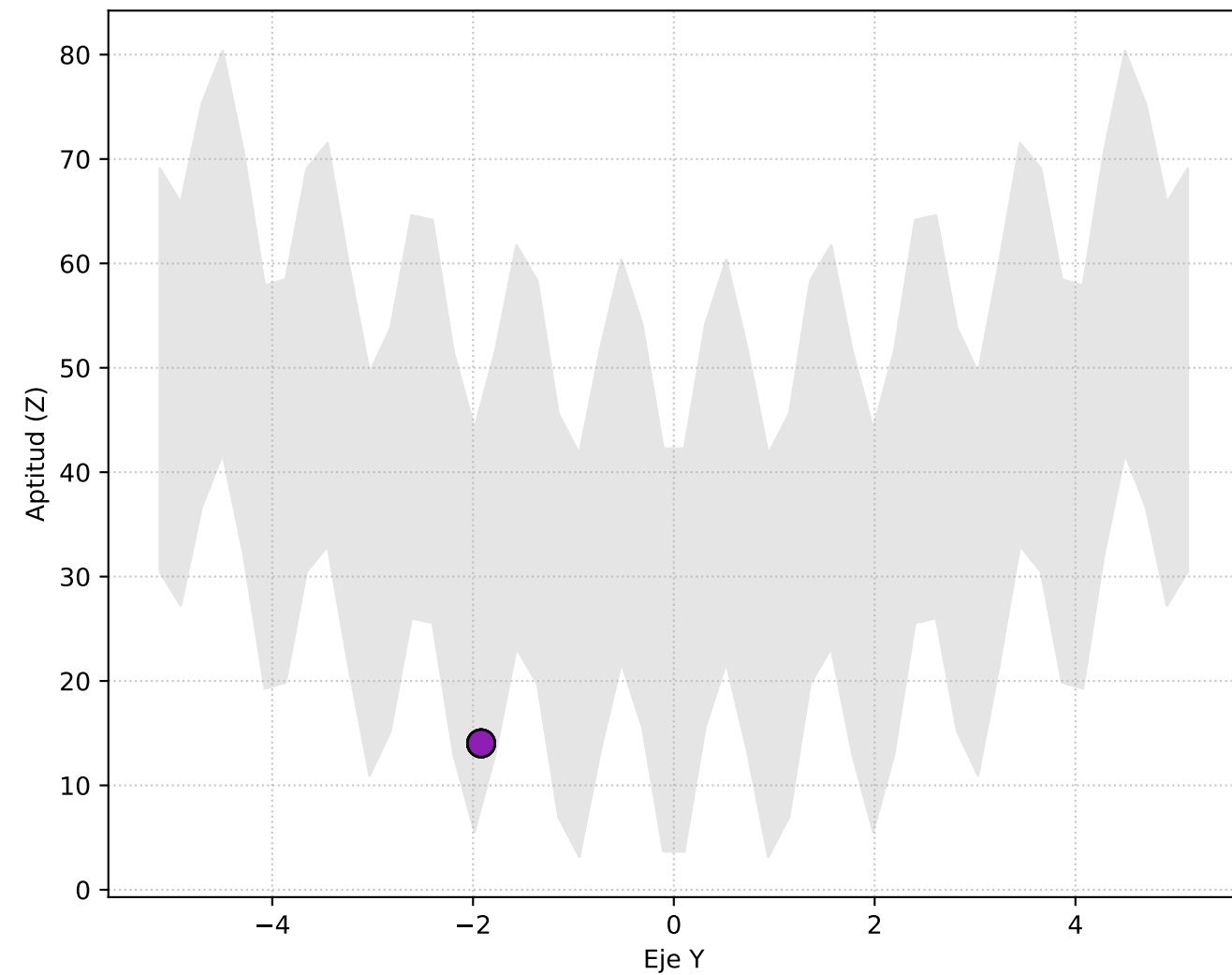
Perspectiva 3D



Vista Lateral (X vs Elevación Z)



Vista Frontal (Y vs Elevación Z)

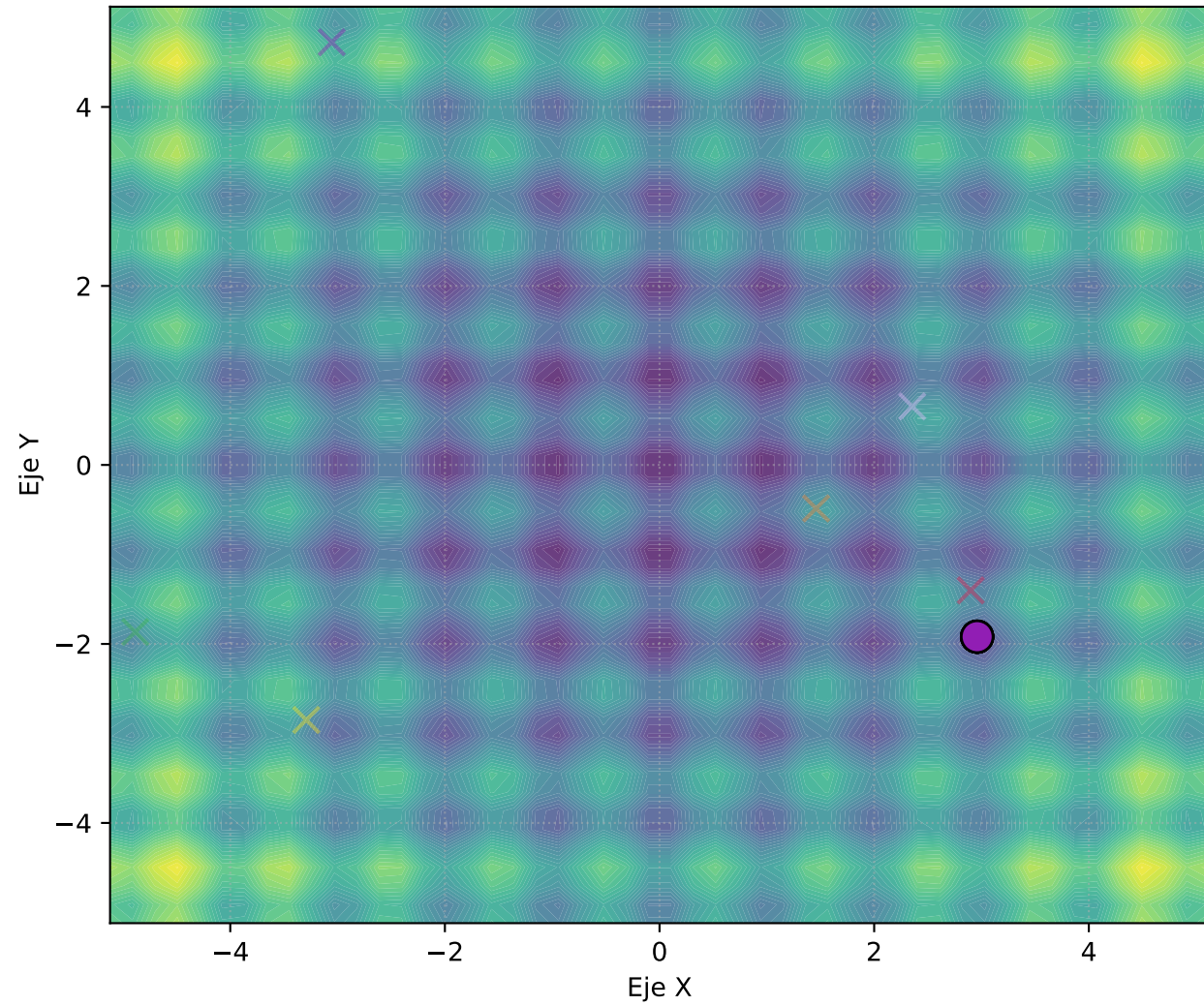


Posición del Mejor Individuo por Salto

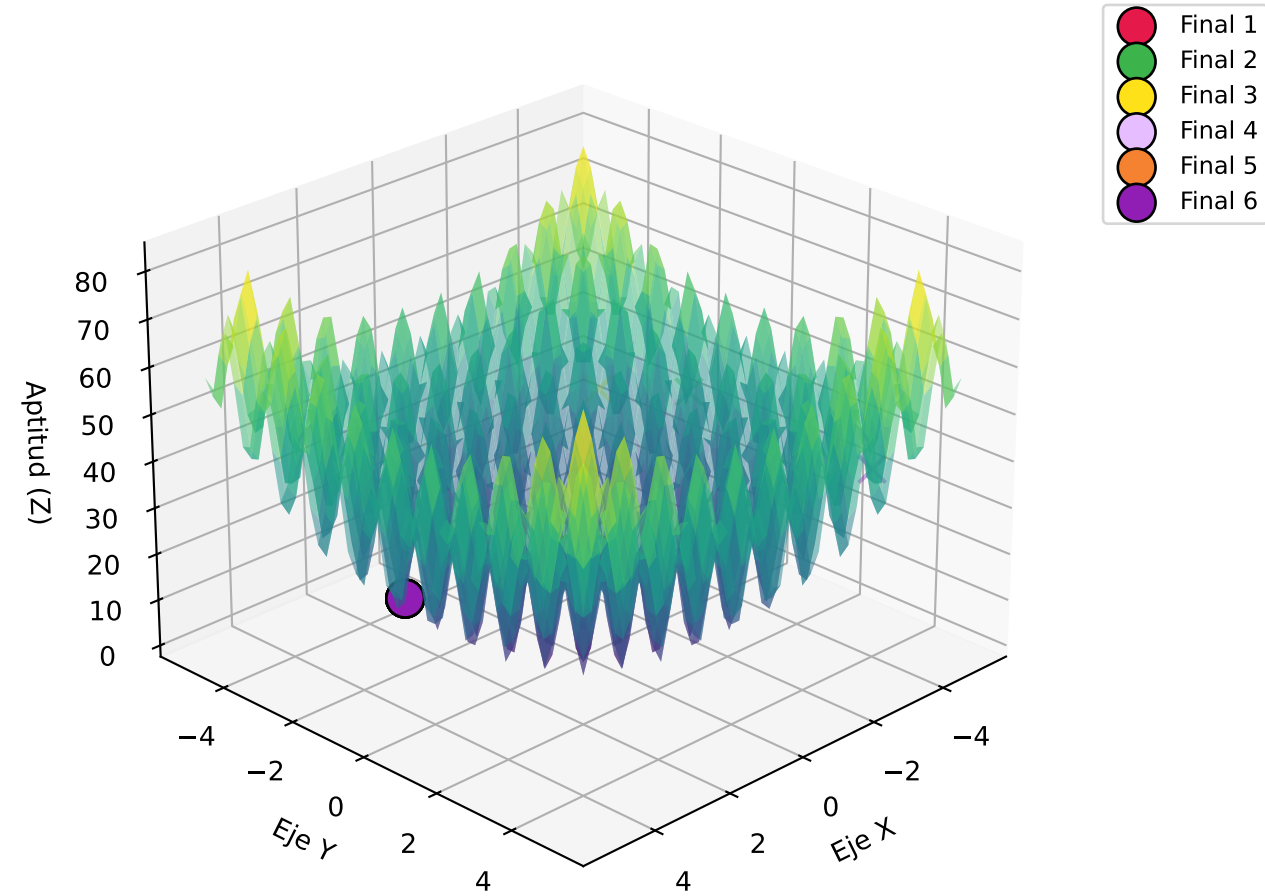
Gen	Mejores Bits	Coordenadas (X, Y)	Aptitud
10	[1, 1, 0, 0, 1, 0, 0, 1, 1, 1, 1, 1, 0, 1, 0, 1, 0, 0, 0, 0, 1, 0, 0]	X: 2.96, Y: -1.91	0.0653
20	[1, 1, 0, 0, 1, 0, 0, 1, 1, 1, 1, 1, 1, 0, 1, 0, 1, 0, 0, 0, 0, 0, 0]	X: 2.96, Y: -1.92	0.0666
30	[1, 1, 0, 0, 1, 0, 0, 1, 1, 1, 1, 1, 1, 0, 1, 0, 1, 0, 0, 0, 0, 0, 0]	X: 2.96, Y: -1.92	0.0666
40	[1, 1, 0, 0, 1, 0, 0, 1, 1, 1, 1, 1, 1, 0, 1, 0, 1, 0, 0, 0, 0, 0, 0]	X: 2.96, Y: -1.92	0.0666
50	[1, 1, 0, 0, 1, 0, 0, 1, 1, 1, 1, 1, 1, 0, 1, 0, 1, 0, 0, 0, 0, 0, 0]	X: 2.96, Y: -1.92	0.0666
60	[1, 1, 0, 0, 1, 0, 0, 1, 1, 1, 1, 1, 1, 0, 1, 0, 1, 0, 0, 0, 0, 0, 0]	X: 2.96, Y: -1.92	0.0666
70	[1, 1, 0, 0, 1, 0, 0, 1, 1, 1, 1, 1, 1, 0, 1, 0, 1, 0, 0, 0, 0, 0, 0]	X: 2.96, Y: -1.92	0.0666
80	[1, 1, 0, 0, 1, 0, 0, 1, 1, 1, 1, 1, 1, 0, 1, 0, 1, 0, 0, 0, 0, 0, 0]	X: 2.96, Y: -1.92	0.0666
90	[1, 1, 0, 0, 1, 0, 0, 1, 1, 1, 1, 1, 1, 0, 1, 0, 1, 0, 0, 0, 0, 0, 0]	X: 2.96, Y: -1.92	0.0666
100	[1, 1, 0, 0, 1, 0, 0, 1, 1, 1, 1, 1, 1, 0, 1, 0, 1, 0, 0, 0, 0, 0, 0]	X: 2.96, Y: -1.92	0.0666

Comparativa: Iniciales (Tache) vs Finales Gen 100 (Círculo)

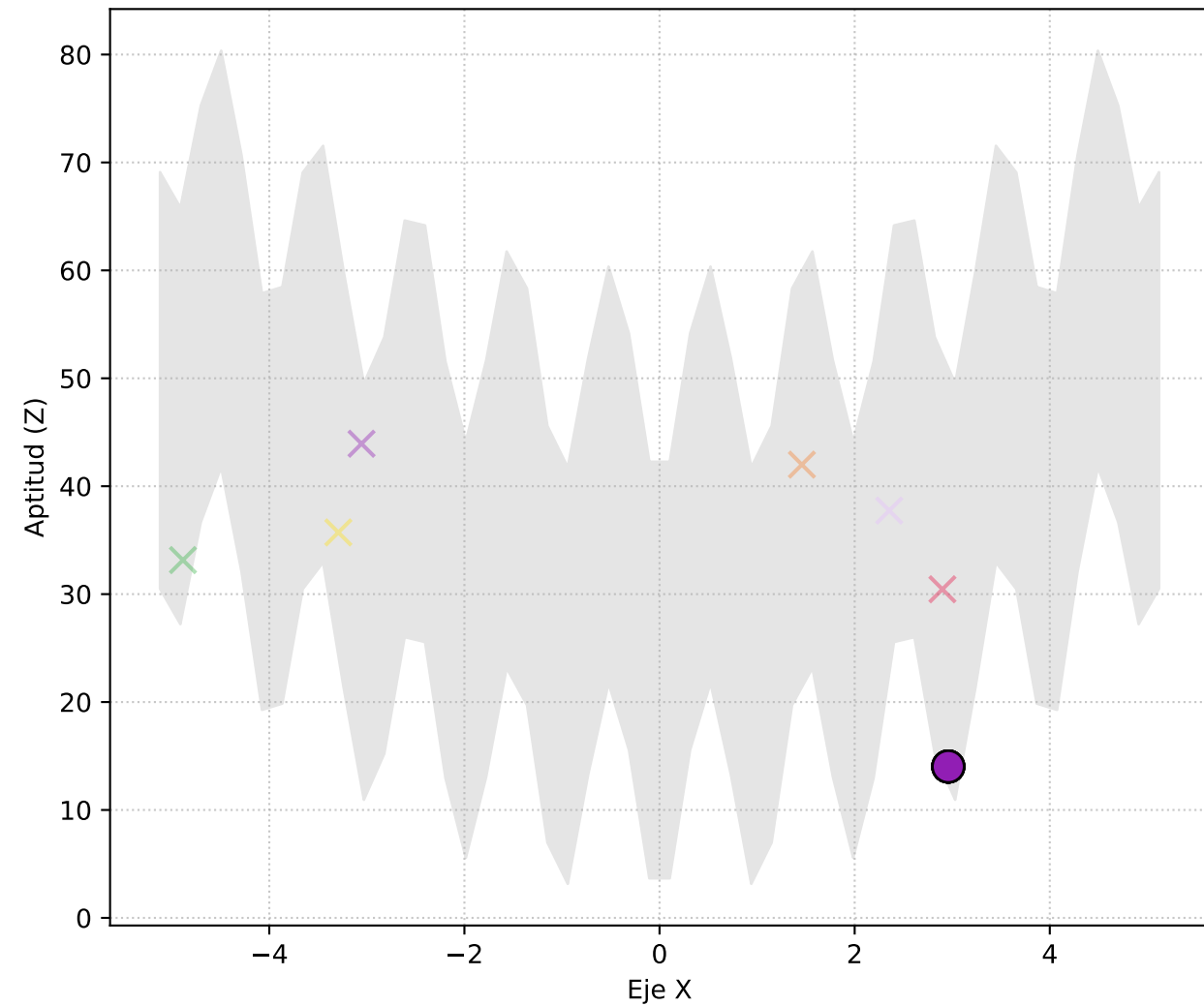
Vista Superior (X, Y)



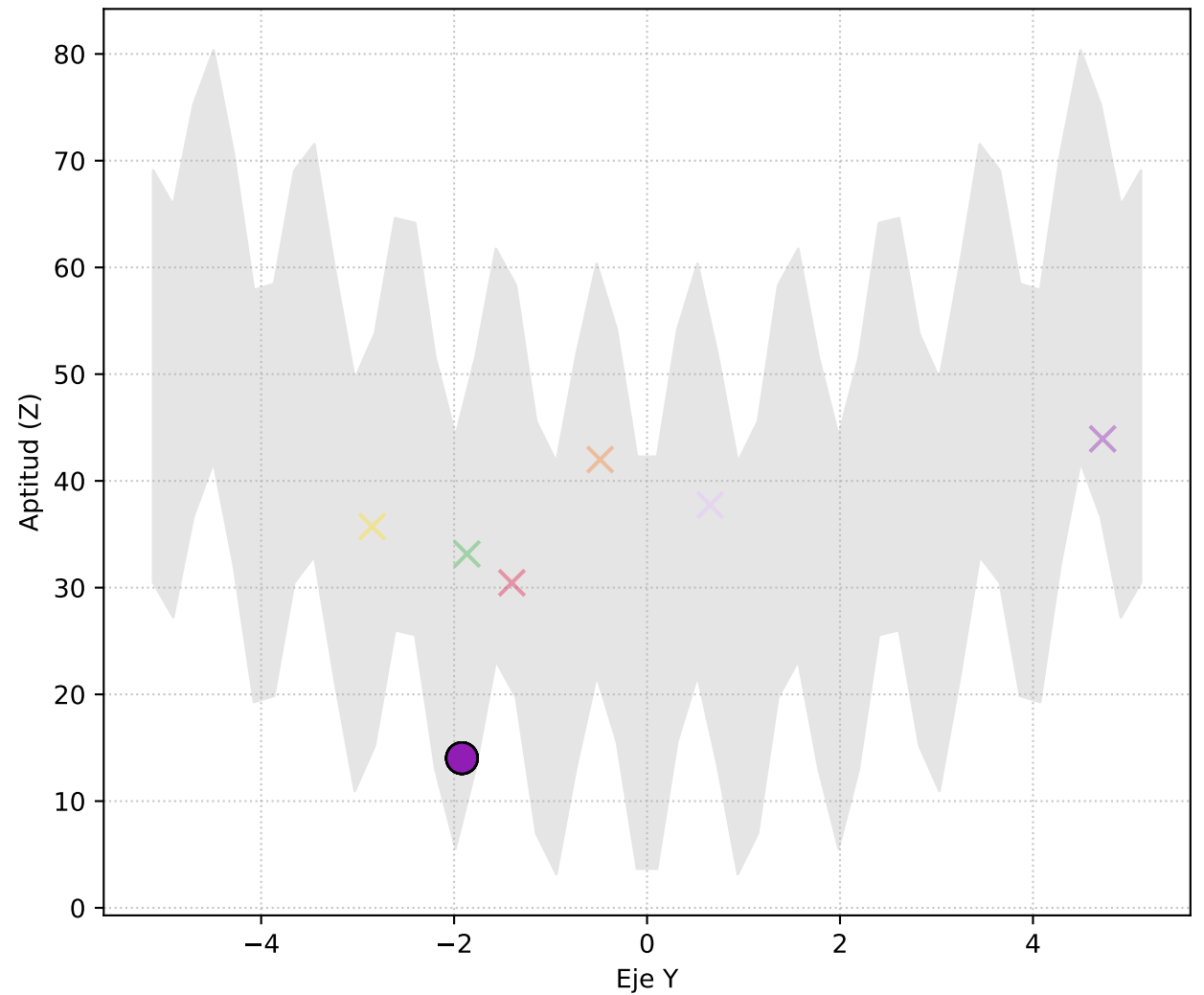
Perspectiva 3D



Vista Lateral (X vs Elevación Z)



Vista Frontal (Y vs Elevación Z)



- Final 1
- Final 2
- Final 3
- Final 4
- Final 5
- Final 6

--- GENERACIÓN 1 ---

1. Evaluación:

Bits: [0, 0, 1, 0, 1, 1, 0, 1, 1, 0, 1, 1, 0, 0, 1, 1, 1, 0, 0, 0, 1, 1, 0, 0] | Dec: (-3.29, -2.85) | Aptitud: 0.0
Bits: [1, 0, 1, 0, 0, 1, 0, 0, 0, 1, 1, 1, 0, 1, 1, 1, 0, 0, 1, 1, 1, 1, 0, 1] | Dec: (1.46, -0.49) | Aptitud: 0.0
Bits: [1, 0, 1, 1, 1, 0, 1, 0, 1, 1, 0, 1, 1, 0, 0, 1, 0, 0, 0, 0, 0, 1, 0, 1] | Dec: (2.35, 0.65) | Aptitud: 0.03
Bits: [0, 1, 0, 1, 1, 1, 1, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 1, 0, 0, 1, 0, 1] | Dec: (-1.36, -3.47) | Aptitud: 0.0
Bits: [1, 1, 1, 0, 1, 1, 1, 1, 0, 1, 0, 1, 1, 1, 0, 1, 0, 1, 0, 1, 0, 0, 0, 1] | Dec: (4.45, 3.40) | Aptitud: 0.01
Bits: [1, 1, 0, 0, 1, 0, 0, 0, 0, 1, 1, 1, 0, 1, 0, 1, 1, 1, 0, 0, 1, 1, 1, 1] | Dec: (2.90, -1.40) | Aptitud: 0.0
Bits: [0, 0, 1, 1, 0, 0, 1, 1, 1, 0, 1, 0, 1, 1, 1, 1, 0, 1, 1, 0, 0, 0, 0, 0] | Dec: (-3.05, 4.72) | Aptitud: 0.0
Bits: [0, 0, 0, 1, 1, 1, 1, 1, 1, 0, 0, 1, 0, 0, 1, 0, 1, 0, 1, 0, 0, 1, 0, 1] | Dec: (-3.86, -3.43) | Aptitud: 0.0
Bits: [1, 1, 1, 1, 0, 0, 0, 1, 0, 0, 0, 1, 1, 0, 1, 0, 1, 1, 1, 1, 1, 0, 0, 1] | Dec: (4.52, 1.90) | Aptitud: 0.02
Bits: [0, 0, 0, 0, 0, 1, 0, 1, 1, 1, 1, 0, 0, 1, 0, 1, 0, 0, 0, 1, 0, 1, 0, 1] | Dec: (-4.88, -1.87) | Aptitud: 0.0

2. Selección:

[0, 0, 0, 0, 0, 1, 0, 1, 1, 1, 1, 0, 0, 1, 0, 1, 0, 0, 0, 1, 0, 1, 0, 1]
[1, 1, 0, 0, 1, 0, 0, 0, 0, 1, 1, 1, 0, 1, 0, 1, 1, 1, 0, 0, 1, 1, 1, 1]
[1, 1, 1, 1, 0, 0, 0, 1, 0, 0, 0, 1, 1, 0, 1, 0, 1, 1, 1, 1, 0, 0, 1]
[1, 0, 1, 0, 0, 1, 0, 0, 0, 1, 1, 1, 0, 1, 1, 1, 0, 0, 1, 1, 1, 1, 0, 1]
[0, 0, 0, 0, 0, 1, 0, 1, 1, 1, 1, 0, 0, 1, 0, 1, 0, 0, 0, 1, 0, 1, 0, 1]

3. Cruza:

Cruzando [0, 0, 0, 0, 0, 1, 0, 1, 1, 1, 1, 0, 0, 1, 0, 1, 0, 0, 0, 1, 0, 1, 0, 1] y [1, 1, 0, 0, 1, 0, 0, 0, 0, 1, 1, 1, 0, 0, 0, 0, 1, 0, 0, 0, 1, 0, 0, 0]
Cruzando [1, 1, 0, 0, 1, 0, 0, 0, 0, 0, 1, 1, 1, 0, 1, 0, 1, 1, 1, 0, 0, 1, 1, 1] y [1, 1, 1, 1, 0, 0, 0, 0, 1, 0, 0, 0, 1, 0, 0, 0, 1, 0, 0, 0, 1, 0, 0, 0]
Cruzando [1, 1, 1, 1, 0, 0, 0, 0, 1, 0, 0, 0, 0, 1, 1, 0, 1, 0, 1, 1, 1, 1, 0, 0, 1] y [1, 0, 1, 0, 0, 1, 0, 0, 0, 1, 0, 0, 0, 1, 0, 0, 0, 0, 1, 0, 0, 0, 1, 0]
Cruzando [1, 0, 1, 0, 0, 1, 0, 0, 0, 1, 1, 1, 0, 1, 1, 1, 0, 0, 0, 1, 1, 1, 1, 0, 1] y [0, 0, 0, 0, 0, 1, 0, 0, 1, 1, 1, 0, 0, 0, 0, 0, 1, 0, 1, 1, 1, 0, 0, 0]
Cruzando [0, 0, 0, 0, 0, 1, 0, 1, 1, 1, 1, 0, 0, 1, 0, 1, 0, 0, 0, 0, 1, 0, 1, 0, 1] y [0, 0, 0, 0, 0, 1, 0, 1, 1, 1, 1, 0, 0, 0, 0, 0, 1, 0, 1, 1, 1, 0, 0, 0]

4. Mutación:

¡Mutación! [1, 1, 0, 0, 1, 0, 0, 1, 0, 0, 0, 1, 1, 0, 1, 0, 1, 1, 1, 1, 1, 0, 0, 1] -> [1, 1, 0, 0, 1, 0, 0, 1, 0, 0, 0, 1, 1, 0, 1, 0, 1, 1, 1, 1, 1, 0, 0, 1]
¡Mutación! [0, 0, 0, 0, 0, 1, 0, 1, 1, 1, 1, 1, 0, 1, 1, 1, 0, 0, 1, 1, 1, 1, 0, 1] -> [0, 0, 0, 0, 0, 1, 0, 1, 1, 1, 1, 0, 1, 1, 1, 1, 0, 0, 1, 1, 1, 1, 0, 1]

--- GENERACIÓN 2 ---

1. Evaluación:

Bits: [0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 1, 1, 1, 0, 1, 0, 1, 1, 1, 0, 0, 1, 1, 1, 1] | Dec: (-4.78, -1.40) | Aptitud: 0.0
Bits: [1, 1, 0, 0, 0, 1, 0, 1, 1, 1, 1, 0, 0, 1, 0, 1, 0, 0, 0, 1, 0, 1, 0, 1] | Dec: (2.80, -1.87) | Aptitud: 0.0
Bits: [1, 1, 0, 0, 1, 0, 0, 1, 0, 0, 0, 1, 0, 0, 1, 0, 1, 1, 1, 1, 1, 0, 0, 1] | Dec: (2.92, -3.22) | Aptitud: 0.0
Bits: [1, 1, 1, 1, 0, 0, 0, 0, 0, 1, 1, 1, 0, 1, 0, 1, 1, 1, 0, 0, 1, 1, 1, 1] | Dec: (4.50, -1.40) | Aptitud: 0.0
Bits: [1, 0, 1, 0, 0, 1, 0, 0, 0, 1, 1, 1, 0, 1, 1, 1, 0, 0, 1, 1, 1, 1, 0, 1] | Dec: (1.46, -0.49) | Aptitud: 0.0
Bits: [1, 1, 1, 1, 0, 0, 0, 1, 0, 0, 0, 1, 1, 0, 1, 0, 1, 1, 1, 1, 1, 0, 0, 1] | Dec: (4.52, 1.90) | Aptitud: 0.02
Bits: [1, 0, 1, 0, 0, 1, 0, 0, 0, 1, 1, 0, 0, 1, 0, 1, 0, 0, 0, 1, 0, 1, 0, 1] | Dec: (1.46, -1.87) | Aptitud: 0.0
Bits: [0, 0, 0, 0, 0, 1, 0, 1, 1, 1, 1, 1, 0, 1, 0, 1, 0, 0, 1, 1, 1, 1, 0, 1] | Dec: (-4.88, -1.77) | Aptitud: 0.0
Bits: [0, 0, 0, 0, 0, 1, 0, 1, 1, 1, 1, 0, 0, 1, 0, 1, 0, 0, 0, 1, 0, 1, 0, 1] | Dec: (-4.88, -1.87) | Aptitud: 0.0
Bits: [0, 0, 0, 0, 0, 1, 0, 1, 1, 1, 1, 0, 0, 1, 0, 1, 0, 0, 0, 1, 0, 1, 0, 1] | Dec: (-4.88, -1.87) | Aptitud: 0.0

2. Selección:

[1, 1, 0, 0, 0, 1, 0, 1, 1, 1, 1, 0, 0, 1, 0, 1, 0, 0, 0, 1, 0, 1, 0, 1]
[1, 1, 0, 0, 0, 1, 0, 1, 1, 1, 1, 0, 0, 1, 0, 1, 0, 0, 0, 1, 0, 1, 0, 1]
[0, 0, 0, 0, 0, 1, 0, 1, 1, 1, 1, 0, 0, 1, 0, 1, 0, 0, 0, 1, 0, 1, 0, 1]
[1, 0, 1, 0, 0, 1, 0, 0, 0, 1, 1, 0, 0, 1, 0, 1, 0, 0, 0, 0, 1, 0, 1, 0, 1]
[1, 1, 0, 0, 1, 0, 0, 1, 0, 0, 0, 1, 0, 0, 0, 1, 0, 1, 1, 1, 1, 1, 0, 0, 1]

3. Cruza:

Cruzando [1, 1, 0, 0, 0, 1, 0, 1, 1, 1, 1, 0, 0, 1, 0, 1, 0, 0, 0, 0, 1, 0, 1, 0, 1] y [1, 1, 0, 0, 0, 1, 0, 1, 1, 1, 1, 0, 0, 1, 0, 1, 0, 1, 1, 1, 0, 1, 1, 1]
Cruzando [1, 1, 0, 0, 0, 1, 0, 1, 1, 1, 1, 0, 0, 1, 0, 1, 0, 0, 0, 0, 1, 0, 1, 0, 1] y [0, 0, 0, 0, 0, 1, 0, 1, 1, 1, 1, 0, 0, 1, 0, 1, 0, 1, 1, 1, 0, 1, 1, 1]
Cruzando [0, 0, 0, 0, 0, 1, 0, 1, 1, 1, 1, 0, 0, 1, 0, 1, 0, 0, 0, 0, 1, 0, 1, 0, 1] y [1, 0, 1, 0, 0, 1, 0, 0, 0, 1, 0, 1, 0, 0, 1, 0, 1, 0, 0, 0, 1, 0, 0, 0]
Cruzando [1, 0, 1, 0, 0, 1, 0, 0, 0, 1, 1, 0, 0, 1, 0, 1, 0, 0, 0, 0, 1, 0, 1, 0, 1] y [1, 1, 0, 0, 1, 0, 0, 0, 1, 0, 1, 1, 1, 1, 1, 1, 0, 0, 0, 1, 0, 1, 1, 1]
Cruzando [1, 1, 0, 0, 1, 0, 0, 1, 0, 0, 0, 0, 1, 0, 0, 0, 1, 0, 1, 1, 1, 1, 0, 0, 1] y [1, 1, 0, 0, 0, 1, 0, 1, 1, 1, 1, 0, 0, 0, 0, 0, 1, 0, 1, 1, 1, 1, 0, 0]

4. Mutación:

¡Mutación! [1, 1, 0, 0, 0, 1, 0, 1, 1, 1, 1, 0, 0, 1, 0, 1, 0, 0, 0, 1, 0, 1, 0, 1] -> [1, 1, 0, 0, 0, 1, 0, 1, 1, 1, 1, 0, 0, 1, 0, 1, 0, 0, 0, 1, 0, 1, 0, 1]
¡Mutación! [1, 0, 1, 0, 0, 1, 0, 0, 0, 1, 1, 0, 0, 0, 1, 0, 1, 0, 0, 0, 1, 0, 1, 0, 1] -> [1, 1, 1, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 1, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0]

Cruzando [1, 1, 0, 0, 1, 0, 0, 1, 1, 1, 1, 1, 0, 1, 0, 1, 0, 0, 0, 0, 0, 0, 0] y [1, 1, 0, 0, 1, 0, 0, 1, 1, 1, 1, 1, 0, 1, 0, 1, 0, 0, 0, 0, 0, 0]

- - - GENERACIÓN 24 - - -

2. Selección:

3. Cruza:

4. Mutación:

--- GENERACIÓN 25 ---

2. Selección:

3. Cruza:

4. Mutación:

Cruzando	[1, 1, 0, 0, 1, 0, 0, 1, 1, 1, 1, 1, 0, 1, 0, 1, 0, 0, 0, 0, 0, 0, 0]	y	[1, 1, 0, 0, 1, 0, 0, 1, 1, 1, 1, 1, 0, 1, 0, 1, 0, 0, 0, 0, 0]
Cruzando	[1, 1, 0, 0, 1, 0, 0, 1, 1, 1, 1, 1, 0, 1, 0, 1, 0, 1, 0, 0, 0]	y	[1, 1, 0, 0, 1, 0, 0, 1, 1, 1, 1, 1, 0, 1, 0, 1, 0, 0, 0, 0, 0]
Cruzando	[1, 1, 0, 0, 1, 0, 0, 1, 1, 1, 1, 1, 0, 1, 0, 1, 0, 1, 0, 0, 0]	y	[1, 1, 0, 0, 1, 0, 0, 1, 1, 1, 1, 1, 0, 1, 0, 1, 0, 0, 0, 0, 0]
Cruzando	[1, 1, 0, 0, 1, 0, 0, 1, 1, 1, 1, 1, 0, 1, 0, 1, 0, 1, 0, 0, 0]	y	[1, 1, 0, 0, 1, 0, 0, 1, 1, 1, 1, 1, 0, 1, 0, 1, 0, 0, 0, 0, 0]

--- GENERACIÓN 59 ---

2. Selección:

3. Cruza:

4. Mutación:

- - - GENERACIÓN 60 - - -

2. Selección:

3. Cruza:

Cruzando [1, 1, 0, 0, 1, 0, 0, 1, 1, 1, 1, 1, 0, 1, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0] y [1, 1, 0, 0, 1, 0, 0, 1, 1, 1, 1, 1, 0, 1, 0, 1, 0, 0, 0, 0, 0, 0]
 Cruzando [1, 1, 0, 0, 1, 0, 0, 1, 1, 1, 1, 1, 0, 1, 0, 1, 0, 1, 0, 0, 0, 0] y [1, 1, 0, 0, 1, 0, 0, 1, 1, 1, 1, 1, 0, 1, 0, 1, 0, 1, 0, 0, 0, 0]
 Cruzando [1, 1, 0, 0, 1, 0, 0, 1, 1, 1, 1, 1, 0, 1, 0, 1, 0, 1, 0, 0, 0, 0] y [1, 1, 0, 0, 1, 0, 0, 1, 1, 1, 1, 1, 0, 1, 0, 1, 0, 1, 0, 0, 0, 0]

4. Mutación:

1. Evaluación:

Bits: [1, 1, 0, 0, 1, 0, 0, 1, 1, 1, 1, 0, 0, 1, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0] | Dec: (2.96, -1.92) | Aptitud: 0.07
Bits: [1, 1, 0, 0, 1, 0, 0, 1, 1, 1, 1, 1, 0, 1, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0] | Dec: (2.96, -1.92) | Aptitud: 0.07
Bits: [1, 1, 0, 0, 1, 0, 0, 1, 1, 1, 1, 1, 0, 1, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0] | Dec: (2.96, -1.92) | Aptitud: 0.07
Bits: [1, 1, 0, 0, 1, 0, 0, 1, 1, 1, 1, 1, 0, 1, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0] | Dec: (2.96, -1.92) | Aptitud: 0.07
Bits: [1, 1, 0, 0, 1, 0, 0, 1, 1, 1, 1, 1, 0, 1, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0] | Dec: (2.96, -1.92) | Aptitud: 0.07
Bits: [1, 1, 0, 0, 1, 0, 0, 1, 1, 1, 1, 1, 0, 1, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0] | Dec: (2.96, -1.92) | Aptitud: 0.07
Bits: [1, 1, 0, 0, 1, 0, 0, 1, 1, 1, 1, 1, 0, 1, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0] | Dec: (2.96, -1.92) | Aptitud: 0.07
Bits: [1, 1, 0, 0, 1, 0, 0, 1, 1, 1, 1, 1, 0, 1, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0] | Dec: (2.96, -1.92) | Aptitud: 0.07
Bits: [1, 1, 0, 0, 1, 0, 0, 1, 1, 1, 1, 1, 0, 1, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0] | Dec: (2.96, -1.92) | Aptitud: 0.07

2. Selección:

[1, 1, 0, 0, 1, 0, 0, 1, 1, 1, 1, 1, 0, 1, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0]
[1, 1, 0, 0, 1, 0, 0, 1, 1, 1, 1, 1, 0, 1, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0]
[1, 1, 0, 0, 1, 0, 0, 1, 1, 1, 1, 1, 0, 1, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0]
[1, 1, 0, 0, 1, 0, 0, 1, 1, 1, 1, 1, 0, 1, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0]
[1, 1, 0, 0, 1, 0, 0, 1, 1, 1, 1, 1, 0, 1, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0]

3. Cruza:

Cruzando [1, 1, 0, 0, 1, 0, 0, 1, 1, 1, 1, 1, 0, 1, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0] y [1, 1, 0, 0, 1, 0, 0, 1, 1, 1, 1, 1, 0, 1, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0]
Cruzando [1, 1, 0, 0, 1, 0, 0, 1, 1, 1, 1, 1, 0, 1, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0] y [1, 1, 0, 0, 1, 0, 0, 1, 1, 1, 1, 1, 0, 1, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0]
Cruzando [1, 1, 0, 0, 1, 0, 0, 1, 1, 1, 1, 1, 0, 1, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0] y [1, 1, 0, 0, 1, 0, 0, 1, 1, 1, 1, 1, 0, 1, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0]
Cruzando [1, 1, 0, 0, 1, 0, 0, 1, 1, 1, 1, 1, 0, 1, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0] y [1, 1, 0, 0, 1, 0, 0, 1, 1, 1, 1, 1, 0, 1, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0]
Cruzando [1, 1, 0, 0, 1, 0, 0, 1, 1, 1, 1, 1, 0, 1, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0] y [1, 1, 0, 0, 1, 0, 0, 1, 1, 1, 1, 1, 0, 1, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0]

4. Mutación:

===== RESULTADO FINAL =====

Bits: [1, 1, 0, 0, 1, 0, 0, 1, 1, 1, 1, 1, 0, 1, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0]

Valor (X, Y): (2.96, -1.92)

Aptitud Alcanzada: 0.07